

STATEMENT OF WORK

Environmental Permitting/Construct New Water Control Structure

PART 1 - PROJECT OVERVIEW

A. NOTICES

1. Project Information

Project Title: Environmental Permitting/Construct New Water Control Structure

Location: U.S. Fish & Wildlife Service
Harrison Lake National Fish Hatchery
11110 Kimages Road
Charles City, VA 23030

Contracting Officer (CO): Khanhthong Moody (Sandy), khanhthong_moody@fws.gov

Project Engineer / Contracting Officer's Representative (COR): Brian Waz, 413-253-8605, brian_waz@fws.gov

FWS Hatchery Manager: Rachel Mair, 804-677-8934, rachel_mair@fws.gov

B. INTRODUCTION

1. Contract Work:

This Contract is to construct a new water control stoplog structure to replace an aging existing water control structure within a man-made canal and to obtain all environmental permitting required to do so. This structure will facilitate improved hydrologic control of the canal to deliver surface water to the fish rearing ponds via two water intake structures located upstream of the water control structure within the man-made canal. During construction, protect surrounding habitats, minimize ecological disturbance, and meet all federal and Commonwealth of Virginia environmental permitting requirements.

The Contractor shall comply with all industry standard best practices and materials.

2. Hatchery Operations:

Hatchery operations will continue during the performance of this contract. The Contracting Officer may require the Contractor to schedule work during a time which would least interfere with the Hatchery operations. The Contractor shall coordinate with the Hatchery Manager and mutually decide on the best time to perform the work and the best method to control the water and turbidity during construction. If it becomes necessary to interrupt any function of the Hatchery, the Contractor shall notify the Hatchery Manager and shall not proceed in such interruptions until such time as the Hatchery Manager makes provisions for and approves such operations. No work shall be performed which could contaminate the water supply or cause harm or death to the existing fish population.

A plan for water control and controlling turbidity during construction must be finalized and agreed upon by all parties before any work can begin.

3. **Anticipated Award Period / Contract Type:**

The Government anticipates a firm-fixed priced award of this contract in September 2025.

4. **Period of Performance:**

The work shall be completed no later than 240 calendar days from the Notice to Proceed issued by the Contracting Officer. The P.O.P. may vary due to environmental permit requirements.

5. **New Water Control Structure Design by McMillen, Inc.:**

The design for the removal and replacement of the water control structure is part of a larger comprehensive design to rehabilitate the water supply system. **Appendix 1** provides the design plans pertinent to the water control structure replacement. **There is no comprehensive As-Built plan available showing all the infrastructure.** A full set of the construction plans produced by McMillen; Inc. will be provided after award of the contract if requested.

PART 2 – SCOPE OF WORK

A. SCOPE

Environmental Compliance and Permitting (Base Bid Item):

(Refer to Plans and Photos in Appendix 1 and 2, respectively)

All work shall comply with applicable federal and Virginia state environmental regulations. The contractor/consultant shall provide items in the list below that are applicable, but not limited to:

- Coordinate with the Virginia Department of Environmental Quality (VA DEQ) and U.S. Army Corps of Engineers (USACE) to obtain:
 - Virginia Water Protection (VWP) Permit (covers state §401 Clean Water Act requirements)
 - Section 404 Nationwide or Individual Permit from USACE Norfolk District
- Prepare a Stormwater Pollution Prevention Plan (SWPPP) as required by the Virginia Stormwater Management Program (VSMP).

Demolition and Replacement of Water Control Structure (Base Bid Item):

(Refer to Plans and Photos in Appendix 1 and 2, respectively)

The Contractor shall provide including, but not limited to:

1. General Requirements:

- Perform all work in accordance with drawings, state and local building codes, and industry standards.
- All aspects of the cast in place concrete work shall conform to ACI 301, “Specifications for Structural Concrete”, and ACI 117.
- Notify Engineer at least 48 hours in advance of intent to place concrete.

- Turbidity and water quality shall be monitored during construction.
- **Materials:**
 - Cement shall comply with ASTM C150, Type I/II and V.
 - Concrete Min. 28-day compressive strength shall be 4,500 psi minimum.
 - Max. Water/Cement Ratio, by weight: 0.42
 - Entrained Air, percent by volume: 3-6
 - Reinforcing shall be new billeted steel complying with ASTM A615, grade 60.
 - Aggregates: Fine ASTM C33, Coarse ASTM C33 (1" max. size for structural concrete).

2. **Water Control Structure Construction:**

- Mobilize equipment and materials to site using designated staging areas and access.
- Install perimeter controls and erosion/sediment BMPs per approved plans, and permits.
- Install Water Control measures. Contractor is responsible for the water control design.
- Demolish existing water control structure.
- Install Forms and all reinforcing rebar per design drawings.
- Pour and cure concrete for new water control structure.
- Minimize construction footprint and prevent turbidity or sedimentation in adjacent waterway.
- Restore construction footprint with native soils and appropriate native seed mix per USFWS and Virginia DWR guidance.
- Stabilize all disturbed areas with erosion control matting or mulch as needed.
- Remove temporary structures, access paths (if not permanent), and debris.

B. ADMINISTRATIVE REQUIREMENTS

1. **Environmental Protection.** All work shall be accomplished in such a manner that all waste materials, pollutants, and potential pollutants shall be completely disposed of in accordance with all applicable Federal, State, and local regulations. At no time shall operations be attempted or permitted where or when such operations present a possibility of fouling the hatchery facility with waste materials, pollutants, or potential pollutants.
2. **Work Site Safety.** The Contractor shall be solely responsible for ensuring that all their on-site activities and equipment conform fully to all standards of the Department of Labor, Occupational Safety and Health Administration (OSHA) 29 CFR 1926 and 1910. Additionally, **the contractor shall adhere to all applicable Federal, State, and Industry standards, Federal and State building codes, and nationally recognized safety codes.** The Contractor shall submit an accident report to the Contracting Officer for any accident that involves lost time by the injured employee regardless of whether professional medical treatment was required or not.
3. **Inadvertent Discovery.** In the unlikely event an inadvertent discovery of cultural material is discovered during the project all work in the vicinity will cease and Robert Frank, Refuge Manager, will be contacted immediately. Refuge staff will take appropriate action to identify and determine significance of the find before work in the area can continue. Work outside the immediate area can continue as planned. Cultural material includes but not limited to, glass, ceramic, bone, brick, etc.

C. PROJECT SUBMITTALS

The following information is required from the Contractor, but not limited to:

Environmental Permitting:

- Complete permit packages and approvals.
- NEPA and ESA compliance documentation.
- Cultural resources coordination records.
- SWPPP and erosion/sediment control plan.

Concrete Water Control Structure:

The following information is required from the Contractor, but not limited to:

- Construction Schedule: Submit the proposed work schedule for approval within 10 days after award of contract prior to any work being started. Itemize each principal component/phase of work and the calendar days of anticipated performance. Submit a revised schedule any time the actual work progress lags the scheduled time by more than 5 calendar days.
- Water Control Design.
- Concrete mix design, furnish that the mix meets specified requirements. Provide items of information as specified under ASTM C94, batch ticket information.
- Concrete slump test results.
- Compressive strength test results.
- Submit manufacturer's information for rebar, and water stops.
- Gradations of aggregates.

Submittals shall be emailed to the following:

Contracting Officer (CO):	TBD
Project Engineer / Contracting Officer's Representative (COR):	Brian Waz, 413-253-8605, brian_waz@fws.gov
FWS Hatchery Manager:	Rachel Mair, 804-677-8934, rachel_mair@fws.gov

1. INSPECTIONS AND ACCEPTANCE

The work performed under this contract are subject to observation and inspection by the USFWS.

2. SITE VISITS

Any work or site visit shall be coordinated with the FWS prior to initiation.

APPENDIX 1:

Water Control Structure Design Plans

	DRAWING INDEX	
95% DESIGN	DWG NO.	DESCRIPTION
GENERAL		
X		COVER SHEET
X	G001	LOCATION MAP AND VICINITY MAP
X	G002	DRAWING INDEX
X	G003	STANDARD ABBREVIATIONS
X	G004	STANDARD SYMBOLS
X	G005	PIPE SCHEDULE
X	G006	VALVE SCHEDULE
X	G007	OVERALL SITE PLAN AND CONTROL
EROSION AND SEDIMENT CONTROL		
X	EC001	STANDARD EROSION AND SEDIMENT CONTROL DETAILS 1
X	EC002	STANDARD EROSION AND SEDIMENT CONTROL DETAILS 2
X	EC100	EROSION AND SEDIMENT CONTROL KEY PLAN
X	EC101	EROSION AND SEDIMENT CONTROL PLAN 1
X	EC102	EROSION AND SEDIMENT CONTROL PLAN 2
X	EC103	EROSION AND SEDIMENT CONTROL PLAN 3
X	EC104	EROSION AND SEDIMENT CONTROL PLAN 4
DEMOLITION		
X	D001	DEMOLITION OVERALL KEY PLAN
X	D201	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 1
X	D202	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 2
X	D203	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 3
X	D204	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 4
X	D205	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 5
X	D206	POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 6
X	D301	LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 1
X	D302	LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 2
X	D303	LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 3
X	D304	LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 4
CIVIL		
X	GC001	CIVIL GENERAL NOTES
X	GC002	STANDARD CIVIL DETAILS 1
X	GC003	STANDARD CIVIL DETAILS 2
X	GC004	STANDARD CIVIL DETAILS 3
X	GC005	STANDARD CIVIL DETAILS 4
X	C001	LAKE WATER SUPPLY CANAL OVERALL SITE KEY PLAN
X	C101	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 1
X	C102	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 2
X	C103	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 3
X	C104	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 4
X	C105	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 5
X	C106	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 6
X	C107	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 7
X	C108	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 8
X	C109	LAKE WATER SUPPLY CANAL PLAN AND PROFILE 9
X	C200	POND DRAIN DISTRIBUTION PIPELINE KEY PLAN
X	C201	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 1
X	C202	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 2
X	C203	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 3

	DRAWING INDEX	
95% DESIGN	DWG NO.	DESCRIPTION
X	C204	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 4
X	C210	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 5
X	C211	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 6
X	C212	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 7
X	C213	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 8
X	C214	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 9
X	C215	POND DRAIN DISTRIBUTION PIPELINE PLAN AND PROFILE 10
X	C300	LAKE WATER DISTRIBUTION PIPELINE KEY PLAN
X	C301	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 1
X	C302	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 2
X	C303	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 3
X	C310	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 4
X	C311	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 5
X	C312	LAKE WATER DISTRIBUTION PIPELINE PLAN AND PROFILE 6
STRUCTURAL		
X	GS001	GENERAL STRUCTURAL NOTES
X	GS002	STANDARD STRUCTURAL DETAILS
X	S001	STRUCTURAL SITE PLAN
X	S100	STOP LOG PLAN
X	S101	STOP LOG SECTIONS AND DETAILS

CONCEPT DESIGN
NOT FOR CONSTRUCTION



VERIFY SCALE	
THIS BAR IS ONE INCH ON ORIGINAL DRAWING	
0"	1"
ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET	

	07/28/23	95% DESIGN SUBMITTAL	RH
	05/05/23	65% DESIGN SUBMITTAL	JB
	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
DRAWING INDEX			G002
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
DRAWING NO.		-----	

ABAN

ABANDONED

ABT

ABOUT

ABV

ABOVE

ABUT

ABUTMENT

AW

ABOVE WATER

ACCEL

ACCELERATE

ADS

ADVANCE DRAINAGE SYSTEM

AHD

AHEAD

ALY

ALLOY

ALT

ALTERNATE

AC

ALTERNATING CURRENT, ASBESTOS CEMENT

AL

ALUMINUM

ASCII

AMERICAN NATIONAL STANDARD CODE FOR INFORMATION

ASWG

AMERICAN STEEL WIRE GAGE

AWG

AMERICAN WIRE GAGE

AHR

ANCHOR

AB

ANCHOR BOLT

&

AND

ASV

ANGLE

APPROX

ANGLE STOP VALVE

ARCH

APPROXIMATE

ACP

ARCHITECTURE

ASPH

ASBESTOS CEMENT PIPE

ACP

ASPHALT

ASSY

ASPHALT CONCRETE PAVEMENT

AUTOCV

ASSEMBLY

ACKTX

AUTOMATIC CHECK VALVE

ADS

AUTOMATIC CIRCUIT EXCHANGE

ATE

AUTOMATIC DOOR SEAL

ATS

AUTOMATIC TEST EQUIPMENT

AUX

AUTOMATIC TRANSFER SWITCH

AUXILIARY

BCS

BEGINNING OF CURVE STATION

BL

BASELINE

BWV

BACKWATER VALVE

BAF

BAFFLE

BALV

BALANCED VOLTAGE

BSP

BALL STOP

BRRG

BARRING

BDIA

BASE DIAMETER

BL

BASE LINE

BATT

BATTEN

BRG

BEARING

BM

BENCH MARK

BEV

BEVEL

BLK

BLOCK

BLKG

BLOCKING

BD

BOARD

BLR

BOILER

BFP

BOILER FEED PUMP

BOT

BOTTOM

BRZ

BRONZE

B.O.

BOTTOM OF

BRKT

BRACKET

BLDG

BUILDING

BLTIN

BUILT-IN

BLDG

BUILDING

BTWLD

BUTT WELD

BTFL

BUTTERFLY

BUZ

BUZZER

BVCS

BEGINNING OF VERTICAL CURVE STATION

BYP

BYPASS

CATV

CABLE TELEVISION

CS

CARBON STEEL

CC.I

CAST IRON

C.I.P.

CAST IN PLACE

CS

CAST STEEL

CB

CATCH BASIN

CTWALK

CATWALK

CTR

CENTER

C.C.

CENTER TO CENTER

CL.

CENTERLINE

CM

CENTIMETER

CID

CHANGE IN DESIGN

CV

CHECK VALUE

CKT

CIRCUIT

CLW

CLARIFIED WATER

CLR

CLEAR

CLSM

CONTROLLED LOW STRENGTH MATERIAL

CO.

CLEANOUT

COR

CORRUGATED

CTG

COATING

CAD

COMPUTER AIDED DESIGN

CONC.

CONCRETE

CCP.

CONCRETE CYLINDER PIPE

CMU.

CONCRETE MASONRY UNIT

CONDUIT

CONN

CONNECTION

CONST

CONSTRUCTION

JT.

CONSTRUCTION JOINT

CONT

CONTINUOUS

COTG

CLEAN OUT TO GRADE

CP

CONTROL POINT

CMP

CORRUGATED METAL PIPE

XBRA

CROSS BRACING

XSECT

CROSS SECTION

XBAR

CROSSBAR

XCONN

CROSS-CONNECTION

CULV

CULVERT

CU FT

CUBIC FEET

CU IN

CUBIC INCH

CUM

CUBIC METER

CUMM

CUYD

CS

CVT

D

d

DMPR

DAMPING

DPG

DEAD LOAD

DL

DECIMAL

DEC

DECK DRAIN VALVE

DDV

DENLUXE

DLX

DENSITY

DENS

DESIGN

DSGN

DESIGN SPECIFICATION

INFORM

DIAGONAL

DIAG

DIAMETER

DIA

DIAPHRAGM

DIAPH

DIGITAL

DGTL

DIMENSION

DIM

DIMMER

DMR

DISABLE

DSBL

DISCHARGE

DISCH

DISCONNECT

DISC

DISCONNECT SWITCH

DS

DISCONNECTING DEVICE

DD

DITTO

DO

DEPARTMENT OF TRANSPORTATION

DOT

DOMESTIC COLD WATER

DCW

DOUBLE

DBL

DRAFTING

DFTG

DRAIN

DR

DRAWING

DWG

DUCTILE IRON

D.I.

EA

EACH

ECS

END OF CURVE STATION

EF

EACH FACE

EW

EACH WAY

E

EAST, EASTING

ELB

ELBOW

ELEC

ELECTRIC

ELECT

ELECTRICAL

ET

ELECTRICAL TIME

ELEM

ELEMENT

EL

ELEVATION

ENCL

ENCLOSURE

ENGY

ENERGY

ENGR

ENGINEER

ENGRG

ENGINEERING

ENV

ENVELOPE

EOR

EDGE OF ROAD

EQL

EQUAL

ERECT

ERECTION

EST

ESTIMATE

EVCS

END OF VERTICAL STATION

EX

EXAMPLE

EXST

EXISTING

EXP

EXPANSION

EXPJT

EXPANSION JOINT

EXT

EXTERIOR

FP

FACEPLATE

FACIL

FACILITY

FCG

FACING

FAC

FACTOR

FED

FEDERAL

FDW

FEEDWATER

FWP

FEEDWATER PUMP

FT.

FEET

FRP

FIBERGLASS REINFORCED PLASTIC

FIG

FIGURE

FG

FINAL GRADE

FNH

FINISH

F.F.

FINISH FLOOR

FABL

FIRE ALARM BELL

FABX

FIRE ALARM BOX

FC

FIRE CONTROL

FCS

FIRE CONTROL SYSTEM

FDR

FIRE DOOR

FH

FIRE HOSE

FHY

FIRE HYDRANT

FW

FIRE WALL

FBCK

FIREBRICK

FP

FIREPLACE

FPRF

FIREPROOF

FWS

FISH AND WILDLIFE SERVICE

FTG

FITTING

FXD

FIXED

FXTR

FIXTURE

FLG

FLANGE

FLR

FLOOR

FD

FLOOR DRAIN

FT

FOOT

F.M.

FLOW METER

F.S.

FLOW SWITCH

FT

FOOT OR FEET

FTG

FOOTING

F.M.

FORCE MAIN

FWD

FORWARD

FDN

FOUNDATION

FR

FRONT

FTK

FUEL TANK

FC TN

FUNCTION

CUBIC MILLIMETER

CUBIC YARD

CYCLE PER SECOND

CULVERT

G

GAL

GPM

GALV.

G.V.

GAR

GASW

GSKT

GAS

GTV

GA.

GEN

GL

GSV

GLV

GLU

GLU-LAN

GOVT

GRD

G.E.

GL

GRTG

GVL

GRL

GRD

GROM

GRV

GRWT

GND

GFI

GDR

GUT

GYP

GWB

GPW

H

H.R.

HNDRL

HDCP

HDL

HDW

HDWD

HAZ

HDR

HTR

HTG

HVY

HD

H

HPRL

HGT

HPT

HWL

MF

HWY

HLDN

HK

HORIZ.

HP

H.B.

HC

HCONN

HGALV

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I

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ISOL

ISLN

ISO

INV

I.E.

IRR.

ISPCW.

FUNL

FURN

FURN

FUBX

FUT

GALLONS

GALLONS PER MINUTE

GALVANIZED

GATE VALVE

GARAGE

GAS WELD

GASKET

GASOLINE

GATE VALVE

GAUGE OR GAGE

GENERATOR

GLASS

GLOBE STOP VALVE

GLOVE VALVE

GLUE-LAMINATED BEAM

GOVERNMENT

GRADE

GRADE ELEVATION

GRADE LINE

GRATING

GRAVEL

GRILLE

GRIND

GROMMET

GROOVE

GROSS WEIGHT

GROUND

GROUND-FAULT

INTERRUPTER

GUARD RAIL

GUTTER

GYP SUM

GYP SUM WALLBOARD

GYP SUM-PLASTER WALL

HANDRAIL

HAND RAIL

HANDICAP

HANDLE

HARDWARE

HARDWOOD

HAZARDOUS

HEADER

HEATER

HEATING

HEAVY

HEAVY DUTY

HIGH

HIGH PRESSURE

RING LOCK

HEIGHT

HIGH POINT

HIGH-WATER LEVEL

HIGHWAY

HOLDDOWN

HOOK

HORIZONTAL

HORSEPOWER

HOSE

HOSE BIB

HOSE CLAMP

HOSE CONNECTOR

HOT-GALVANIZE

HOUSING

HYBRID

IGNITION

ILLUMINATE

IMPACT

INCH

INCH PER SECOND

INCLOSURE

INCLUDE

INFLATABLE

INFORMATION

INJECT

INLET

INPUT-OUTPUT

INSIDE

INSIDE DIAMETER

INSIDE RADIUS

INSTALL

INSTALLATION

INSTRUMENT

INSULATION

INTAKE

INTERFACE

INTERIOR

INTERNATIONAL PIPE STANDARD

IRON PIPE

IRON PIPE SIZE

IRON PIPE THREAD

ISLAND

ISOLATE

ISOLATION

ISOMETRIC

INVERT

INVERT ELEVATION

IRRIGATION

IDAHO STANDARDS FOR PUBLIC WORKS CINSTRUCTION

J

JKT

JKSCR

JT

JCT

J

K

K

KPL

KP

KM

L

LAD

LDG

LDMK

LGE

LATCH

LATL

LAT

LAU

LBK

LH

LS

LHDR

LHS

LG

LVL

LVR

LFT

LT

LTSW

LTG

L.F.

LNG

LIQ

LL

LOC.

LKWASH

LKNT

LKSCR

LG.

LLH

LLV

LVR

LP

LW

LWR

LWST

LBR

M

MACH

MST

MAG

MCT

MC

MF

MAINT

MH

MC

MFR

MASY.

MSW

MATL

MAX

MHW

MHHW.

MLW.

MLLW.

MSL

MECH

MJ

MED.

MET

METD

METF

METR

M

MI

MPH

MM

MIN

MISC

MOD

MOT

MT

MTD

MTG

MVG

N

NATL

NECS

NFH

NWR

NWRC

NWG

NKL

NICOP

NCA

NS

NP

NIP

NOM

JACKET

JACKSCREW

JOINT

JUNCTION

JUNCTION BOX

KIP

KICK PLATE

KICKPIPE

KILOMETER

LADDER

LANDING

LANDMARK

LARGE

LATCH

LATERAL

LATITUDE

LAUNDRY

LEFT BANK

LEFT HAND

LEFT SIDE

LEFT-HAND DRIVE

LEFT-HAND SIDE

LENGTH or LONG

LEVEL

LEVER

LIFTING

LIGHT

LIGHT SWITCH

LIGHTING

LINEAL FEET

LINING

LIQUID

LIVE LOAD

LOCATION

LOCK WASHER

LOCKNUT

LOCKSCREW

LONG

LONG LEG HORIZONTAL

LONG LEG VERTICAL

LOUVER

LOW POINT

LAKE WATER

LOWER

LOWEST

LUMBER

MACHINE

MACHINE STEEL

MAGNET

MAIN CONTROL TANK

MAIN COOLANT

MAIN FEED

MAINTENANCE

MANHOLE

MANHOLE COVER

MANUFACTURE

MASONRY

MASTER SWITCH

MATERIAL

MAXIMUM

MEAN HIGH WATER

MEAN HIGHER HIGH WATER

MEAN LOW WATER

MEAN LOWER LOW WATER

MEAN SEA LEVEL

MECHANICAL

MECHANICAL JOINT

MEDIUM

METAL

METAL DOOR

METAL FLASHING

METAL ROOF

METER

MILE

MILES PER HOUR

MILLIMETER

MINIMUM

MISCELLANEOUS

MODEL

MOTOR

MOUNT

MOUNTED

MOUNTING

MOVING

NATIONAL

NATIONAL ELECTRICAL CODE

NATIONAL FISH HATCHERY

NATIONAL WILDLIFE REFUGE

NATIONAL WILDLIFE REFUGE

COMPLEX

NATIONAL WIRE GAGE

NICKEL

NICKEL COPPER

NICKEL COPPER ALLOY

NICKEL STEEL

NICKEL-PLATED

NIPPLE

NOMINAL

NC

NO

N

N/A

NIC

NTS

NOZ

NO. OR #

NYL

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OBJ

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OPTL

ORF

OPP

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OR

OVFL

O.H.

OHD.

ONLY

OVS

P

PNT

PNL

PRKG

PATT

PVMNT.

PEN

PER

PERF

PERP

PH

PI

PIP

PLCS.

PL

PPLN

PLMB

POC

PT

P.T.

PVC

PE

LB(S)

PSF

PSI

PWR

PEMB

PJF

PRCST

P.T.

PF

PRGM

PROJ

PPP

PS

PDISCH

PRG

PRNG

PRFCN

PB

Q

QVT

QTY

R

RADN

R

RLG

RR

RY

REAC

RDOUT

RDCR

REF

RCB

RCP

REINF.

RFS

RLY

REPL

REQ

REQ

REQD

REV

RBK

RH

NORMALLY CLOSED

NORMALLY OPEN

NORTH, NORTHING

NOT APPLICABLE

NOT IN CONTRACT

NOT TO SCALE

NOZZLE

NUMBER

NYLON

OBJECT

ON CENTER

OPENING

OPTICAL

OPTIONAL

ORIFICE

OPPOSITE

OUTLET

OUTSIDE CIRCUMFERENCE

OUTSIDE DIAMETER

OUTSIDE FACE

OUTSIDE RADIUS

OVERFLOW

OVERHEAD

OVERHEAD DOOR

OVERLOAD RELAY

OVERSIZE

PAINT

PANEL

PARKING

PATTERN

PAVEMENT

PENETRATE

PENETRATION

PERFORATED

PERPENDICULAR

PHRASE

POINT OF INTERSECTION

PLASTIC IRRIGATION PIPE

PLACES

PLATE

PIPELINE

PLUMBING

POINT OF CONNECTION

POINT

POLLUTION ABATEMENT

POLYVINYL CHLORIDE

POLYETHYLENE

POUND(S)

POUNDS PER SQUARE FOOT

POUNDS PER SQUARE INCH

POWER

PRE-ENGINEERED METAL BUILDING

PRE-FORMED JOINT FILLER

PRECAST

PRESSURE TREATED

PROFILE

PROGRAM

PROJECT

PULL AND PUSH PLATE

PULL SWITCH

PUMP DISCHARGE

PURGE

PURGING

PURIFICATION

PUSH BUTTON

QUALITY VERIFICATION TESTING

QUANTITY

RADIATION

RADIUS

RAILING

RAILROAD

RAILWAY

REACTOR

READOUT

REDUCER

REFERENCE

REINFORCED CONC.

BOX CULVERT

REINFORCED CONCRETE

PIPE

REINFORCED

RADIAL FLOW SEPARATOR

RELAY

REPLACE

REQUEST

REQUIRE

REQUIRED

REVISION

RIGHT BANK

RIGHT HAND

RS

RD.

RFG

ROT

RGH

RND

RINSUL

RPP

RSTPF

RIGHT SIDE

ROAD

ROOFING

ROTATE

ROUGH

ROUND

RUBBER INSULATION

REINFORCED POLYPROPYLENE

RUSTPROOF

S

SANS

SDL

SV

SC

SCH.

SCD

SCRGT

SL

SEC

SECT

SW

SHTHG

SHT.

SHLN

SHW

SDG

SIM

SGL

SPH

SLT

SLV

SLD

SLP

SND

S

SPA.

SPCR

SPEC

SQ

SQCM

SQFT

SQIN

SQKM

S.S.

STWY

STD

STDF

STA

STL

STIFF

SD

ST

STGR

STRL

STRUCT

SMTK

SPLY

SPRT

SURF

S.W.

SURV

SW

SYM

SYMM

T

TEMP

TBM

THK

THKR

THKNS

TNR

3WAY

TMBR

T.F.

T.S.

T & G

TC

T.O.

T.O.C.

TOS

TOW

TLLD

XFMR

TRTD

TRB

TRK

TR

TS

TNL

TYP

TEMPORARY

TEMPORARY BENCH MARK

THICK

THICKER

THICKNESS

THINNER

THREE-WAY

TIMBER

TOE OF FILL

TOE OF SLOPE

TONGUE AND GROOVE

TOP CHORD

TOP OF

TOP OF CONCRETE

TOP OF SLAB

TOP OF WALL

TOTAL LOAD

TRANSFORMER

TREATED

TREBLE

TRUNK

TRUSS

TUBE SHEET

TUNNEL

TYPICAL

U

UHMWPE

UV

UGND

UNFIN

UN

US

USFWS

ULTRA HIGH MOLECULAR

WEIGHT POLYETHYLENE

ULTRAVIOLET

UNDERGROUND

UNFINISHED

UNION

UNITED STATES

UNITED STATES FISH

AND WILDLIFE SERVICE

UNKNOWN

UNLESS NOTE OTHERWISE

UPPER

URINAL

USE AS REQUIRED

USED WITH

UTILITY

V

VAC

VAL

V

VB

VS

VAR

VEL

VERIF

VERT

VSBL

V

VCT

VDET

VPI

VRL

VOL

VACUUM

VALLEY

VALVE

VALVE BOX

VAPOR SEAL

VARIABLE

VELOCITY

VERTICAL

VISIBLE

VOLT

VOLTAGE CONTROL TRANSFER

VOLTAGE DETECTOR

VERTICAL POINT OF INTERSECTION

VOLTAGE RELAY

VOLUME

W

WASH

WTR

WR

WS

WV

WLB

WSHR

WP

WTR

WCHR

WC

WCR

WD

WH

WP

WS

WCLD

WL

WTRPRF

WWHL

WAT

WSL

WSEL

WTHPRF

WBG

WT

WLD

WWF

W

WD

WDO

WB

WG

WM

WW

W/

W/O

WD

WB

WBL

WD

WDF

WDP

WT

WB

WP

WKS

WASHINGTON

WATER

WALL RECEPTACLE

WATER SURFACE

WALL VENT

WALLBOARD

WASHER

WASTE PIPE

WATER

WATER CHILLER

WATER CLOSET

WATER COOLER

WATER DRAIN

WATER HEATER

WATER PUMP

WATER STOP

WATER-COOLED

WATERLINE

WATERPROOF

WATERWHEEL

WATT

WATER SEAL

WATER SURFACE ELEVATION

WEATHERPROOF

WEBBING

WEIGHT

WELDED

WELDED WIRE FABRIC

WEST, WESTING

WIDTH

WINDOW

WINDOW DIMENSION

WIRE GAGE

WIRE MESH

WIREWAY, WELL WATER

WITH

WITHOUT

WOOD

WOOD BASE

WOOD BLOCKING

WOOD DOOR

WOOD DOOR AND FRAME

WOOD PANEL

WOOD THRESHOLD

WORKBENCH

WORKING POINT

WORKSHOP

Y

YD

YARD

VERIFY SCALE

THIS BAR IS ONE INCH ON ORIGINAL DRAWING

0"1"

ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET

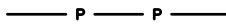
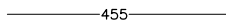
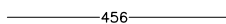
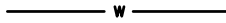
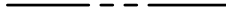
McMillen

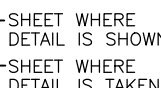
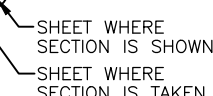
CONCEPT DESIGN

NOT FOR CONSTRUCTION

 07/28/23	95% DESIGN SUBMITTAL	RH
 05/05/23	65% DESIGN SUBMITTAL	JB
 10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION		
STANDARD ABBREVIATIONS		
DESIGNED	DRAWN	CHECKED
RH	FH	JB
DATE	DRAWING NO.	
07/28/23	----	

G003

SITE PLAN SYMBOLS		HATCH SYMBOLS		SITE PLAN LINETYPES		GENERAL NOTES:			
	CONIFER TREE: FIR, SPRUCE, LARCH OR PINE, 8" DIAMETER OR LARGER		GRAVEL/ROCK, TYPE AS NOTED (PLAN/SECTION)		FENCE LINE	1. ABBREVIATIONS ON SHEET G003 APPLY TO THE ENTIRE SET OF CONTRACT DRAWINGS. 2. LISTING OF ABBREVIATIONS ON SHEET G003 DOES NOT IMPLY ALL ABBREVIATIONS ARE USED IN THE CONTRACT DRAWINGS. 3. ABBREVIATIONS SHOWN ON SHEET G003 INCLUDE VARIATIONS OF THE WORD. FOR EXAMPLE, "MOD" MAY MEAN MODIFY OR MODIFICATION; "INC" MAY MEAN INCLUDED OR INCLUDING; "REINF" MAY MEAN EITHER REINFORCE OR REINFORCING. 4. SCREENING OR SHADING OF WORK IS USED TO INDICATE EXISTING COMPONENTS OR TO DE-EMPHASIZE PROPOSED IMPROVEMENTS TO HIGHLIGHT SELECTED TRADE WORK. REFER TO CONTEXT OF EACH SHEET FOR USAGE. 5. NOT ALL SYMBOLS ON THIS SHEET ARE NECESSARILY USED. THIS IS A STANDARD DRAWING SHOWING COMMON SYMBOLS.			
	DECIDUOUS TREE: COTTONWOOD, HAWTHORN, ASPEN, 8" DIAMETER OR LARGER		BED ROCK, RIPRAP		OVERHEAD POWER				
	MANHOLE		EXISTING GRADE (SECTION)		MAJOR CONTOUR				
	ELECTRIC BOX		NEW SOIL (SECTION)		MINOR CONTOUR				
	STORM DRAIN MANHOLE		CONCRETE (SECTION/PLAN)		DITCH CENTERLINE				
	SURVEY CONTROL POINT		SAND, GROUT (PLAN/SECTION)		EXIST TOE OF SLOPE				
	SURVEY MONUMENT		STEEL (SECTION)		EXIST TOP OF BANK				
	GATE VALVE		WOOD, SIZE/TYPE AS NOTED (PLAN)		TOE OF SLOPE				
	TRAFFIC SIGN		WOOD, SIZE/TYPE AS NOTED (SECTION)		TOP OF BANK				
	WATER VALVE		STAGING AREA		EDGE OF PAVEMENT				
	EXIST CATCH BASIN INLET		GRANULAR SUBGRADE (SECTION)		EXIST EDGE OF WATER				
	ENTRY GATE		DEMOLITION/REMOVAL		WATTLE				
	UTILITY VAULT				SILT FENCE				
	LIGHT POLE				CONSTRUCTION FENCE				
	TREE				STRAW WATTLE				
	FIBER OPTICS PEDESTAL				ORDINARY HIGH WATER LINE				
					SECTION LINE				
					RAILROAD RIGHT-OF-WAY				
					CENTERLINE				
					ASPHALT SAWCUT				
					PROJECT LIMITS				
					EXIST EDGE OF PAVEMENT				
					EXIST FENCE LINE				
					EXIST CULVERT				
					EXIST FIBER OPTIC				
					EXIST OVERHEAD POWER				
					EXIST UNDERGROUND POWER				
					EXIST ROAD BUMP				
					EXIST EDGE OF GRAVEL				
					EXIST DRAIN PIPE				
					EXIST WELL WATER				
					EXIST WATER SUPPLY				
					LAKE WATER SUPPLY PIPE				
					DRAIN PIPE				

DETAILING CONVENTIONS	
DETAILS:  DETAIL NUMBER  SHEET WHERE DETAIL IS SHOWN SHEET WHERE DETAIL IS TAKEN	SECTIONS AND VIEWS:  SECTION LETTER  SHEET WHERE SECTION IS SHOWN SHEET WHERE SECTION IS TAKEN

CONCEPT DESIGN
NOT FOR CONSTRUCTION



VERIFY SCALE

THIS BAR IS ONE INCH ON ORIGINAL DRAWING

0" 1"

ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET

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REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
STANDARD SYMBOLS			G004
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
DRAWING NO.			

FLUID ABBREVIATION	FUNCTION			FIELD TEST REQUIREMENTS (SEE NOTE 3 AND NOTE 4)		
	THIS LIST MAY INCLUDE FLUIDS NOT USED IN THIS PROJECT	BURIED PIPING (SEE NOTE 9)		MINIMUM TEST PRESSURE PSI	TEST MEDIUM	LEAKAGE ALLOWANCE (SEE NOTE 2)
	(* SEE NOTE 5)	3" DIA AND SMALLER	4" DIA AND LARGER			
COMMONLY USED FUNCTIONS						
LW	LAKE WATER	31	31	75	WATER	(A)
DR	POND DRAIN	31	31	75	WATER	(A)

PIPING MATERIAL SCHEDULE (SEE NOTE 1)			
GROUP NO.	PIPE MATERIAL	FITTINGS / JOINTS	LININGS AND COATINGS
31	HIGH DENSITY POLYETHYLENE (HDPE) ASTM D3350 – DR 21 IPS PE 4710 MATERIAL, UNLESS OTHERWISE NOTED. (NOTE 8)	HDPE THERMAL BUTT WELD; FLANGE CONNECTIONS AT ALL VALVES AND TRANSITIONS.	NOT APPLICABLE

TYPICAL PIPE DESIGNATION:

2" UW (24)

PIPE DIAMETER

MATERIAL GROUP NUMBER

FLUID ABBREVIATION

NOTE 1

ALTHOUGH SEVERAL PIPE MATERIAL GROUPS MAY BE LISTED ON THIS SHEET FOR A GIVEN FLUID SERVICE, CONTRACTOR SHALL PROVIDE ONLY THE PIPE MATERIAL GROUP SHOWN ON THE DRAWINGS AND SPECIFIED FOR THAT FLUID SERVICE.

NOTE 2

LEAKAGE ALLOWANCE IS AS FOLLOWS
A. PIPES SO DESIGNATED SHALL SHOW ZERO LEAKAGE.
B. PIPES SO DESIGNATED SHALL SHOW ZERO LEAKAGE FOR UNBURIED PIPE AND NOT MORE THAN 0.02 GALLON PER HOUR PER INCH DIAMETER PER 100 FEET OF BURIED PIPE.
C. PIPES SO DESIGNATED SHALL NOT SHOW A LEAKAGE OF MORE THAN 0.15 GALLON PER HOUR PER INCH OF DIAMETER PER 100 FEET OF PIPE.
D. PIPES SO DESIGNATED SHALL NOT SHOW A LOSS OF PRESSURE OF MORE THAN 5 PERCENT.
E. PIPE SO DESIGNATED SHALL NOT SHOW A LOSS OF VACUUM OF MORE THAN 4 INCHES MERCURY COLUMN.

NOTE 3

FOR FIELD TEST PROCEDURES AND ADDITIONAL TEST REQUIREMENTS, SEE PIPING SECTION OF SPECIFICATIONS.

NOTE 4

STATIC WATER TEST WITH SURFACE 5 FEET ABOVE HIGH POINT OF PIPE.

NOTE 5

CHANGE IN PIPING MATERIAL GROUP NUMBER IS INDICATED THUS: —◆—

NOTE 6

FOR FULL PIPE LINING AND COATING REQUIREMENTS, SEE SPECIFICATIONS.

NOTE 7

EXPOSED OUTDOOR PIPING SHALL BE PAINTED IN ACCORDANCE WITH SPECIFICATIONS. COLORS TO BE SELECTED BY OWNER.

NOTE 8

FOR HDPE PIPING THE SIZE OF PIPE SHOWN ON DRAWING CALL-OUTS SHALL BE THE MINIMUM INSIDE DIAMETER. PIPE WALL THICKNESS SHALL BE PER DR RATING REQUIREMENT.

NOTE 9

FOR FULL PIPE LINING AND COATING REQUIREMENTS, SEE SPECIFICATIONS.

CONCEPT DESIGN
NOT FOR CONSTRUCTION



VERIFY SCALE

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0"—————1"

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	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
PIPE SCHEDULE			G005
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
		DRAWING NO.	----

VALVE SCHEDULE								
EQUIPMENT NUMBER	LOCATION	FLUID / SERVICE	TYPE	DIAMETER (IN)	PRESSURE CLASS (AWWA/ANSI)	ENDS	ACTUATOR TYPE (NORMAL POSITION)	BODY AND DISC MATERIAL
V-E11	POND E1	LAKE WATER / KETTLE SUPPLY	GATE	6	75B OR 150B	ALPHA	SQUARE NUT FOR TEE HANDLE (OPEN)	CAST IRON
V-F11	POND F1	LAKE WATER / KETTLE SUPPLY	GATE	6	75B OR 150B	ALPHA	SQUARE NUT FOR TEE HANDLE (OPEN)	CAST IRON
V-F21	POND F2	LAKE WATER / KETTLE SUPPLY	GATE	6	75B OR 150B	ALPHA	SQUARE NUT FOR TEE HANDLE (OPEN)	CAST IRON
V-G11	POND G1	LAKE WATER / KETTLE SUPPLY	GATE	6	75B OR 150B	ALPHA	SQUARE NUT FOR TEE HANDLE (OPEN)	CAST IRON
V-G21	POND G2	LAKE WATER / KETTLE SUPPLY	GATE	6	75B OR 150B	ALPHA	SQUARE NUT FOR TEE HANDLE (OPEN)	CAST IRON
V-DR1	HERRING CREEK	POND DRAIN WATER	TIDEFLEX CHECK VALVE SERIES TF-1	10	125# OR 150#	304 SST CLAMP	N/A	EPDM
V-DR2	HERRING CREEK	POND DRAIN WATER	TIDEFLEX CHECK VALVE SERIES TF-1	10	125# OR 150#	304 SST CLAMP	N/A	EPDM

CONCEPT DESIGN
NOT FOR CONSTRUCTION

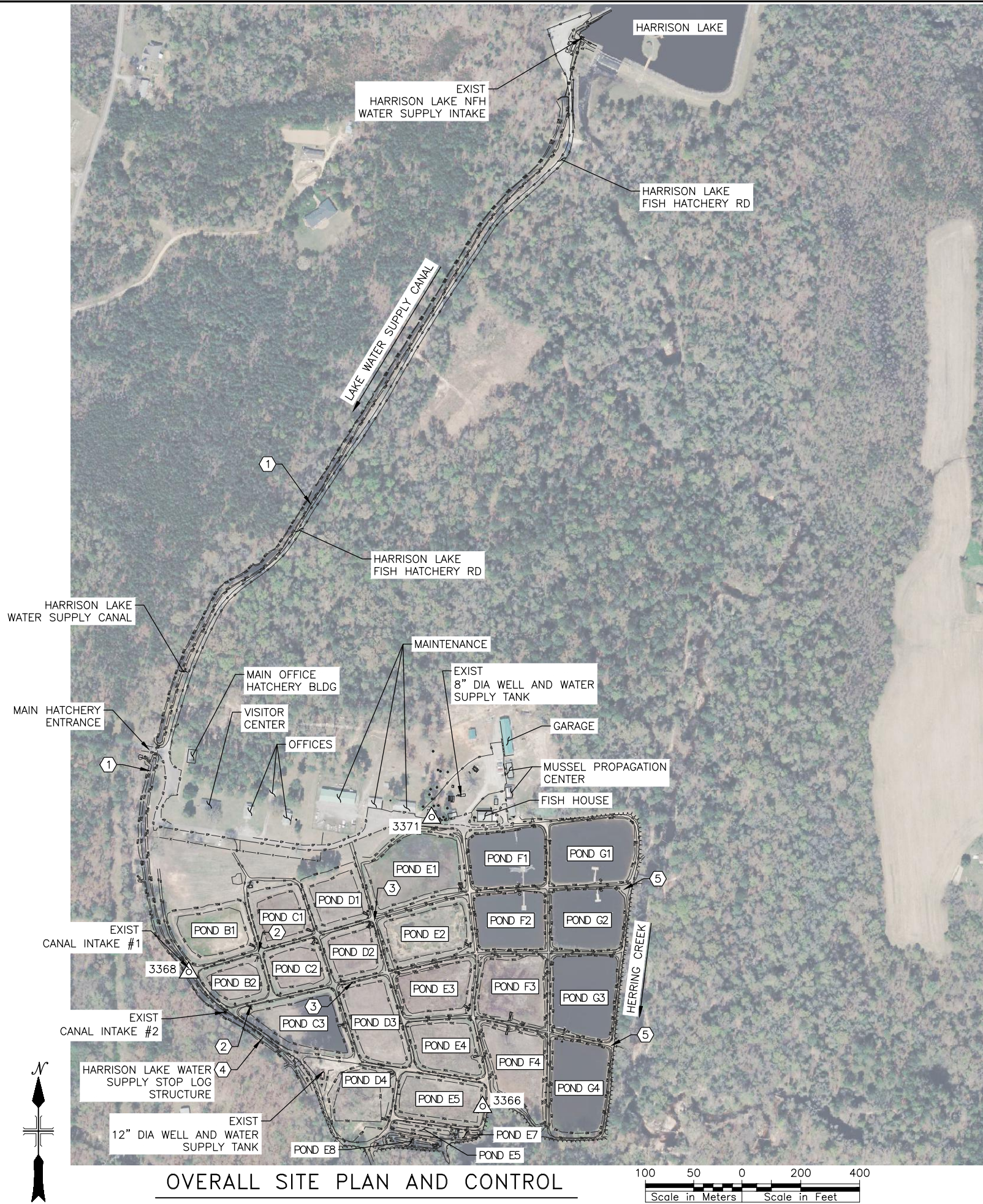


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REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
VALVE SCHEDULE			G006
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
DRAWING NO.		----	



- SHEET NOTES:
1. VERTICAL DATUM: NORTH AMERICAN VERTICAL DATUM OF 1988 (NAVD88) MEAN SEA LEVEL (MSL), US FOOT.
 2. HORIZONTAL DATUM: NAD83 VIRGINIA STATE PLANE, SOUTH ZONE, US FOOT
 3. TOPOGRAPHIC SURVEY COMPLETED BY COLLIERS ENGINEERING & DESIGN, AUGUST 2022 WITH PROJECT BENCH MARKS AS INDICATED.

- MAJOR WORK ITEMS:
1. REPAIR EXISTING CANAL BANKS AND DREDGE CHANNEL WHERE INDICATED ON DRAWINGS.
 2. DEMOLISH EXISTING CANAL SUPPLY PIPELINES FROM INTAKE #1 AND INTAKE #2. EXISTING PIPELINES ARE WROUGHT IRON OR ASBESTOS CEMENT (AC) PIPE. CONTRACTOR SHALL VERIFY LOCATION OF AC PIPE PRIOR TO DEMOLITION AND SHALL TAKE NECESSARY PRECAUTIONS DURING DEMOLITION TO HANDLE AND DISPOSE OF AC PIPE.
 3. PROVIDE NEW LAKE WATER SUPPLY PIPING FROM INTAKE #1 AND INTAKE #2 TO HATCHERY PONDS.
 4. DEMOLISH EXISTING STOP LOG STRUCTURE AND CONSTRUCT NEW SUPPLY CANAL STOP LOG STRUCTURE.
 5. PROVIDE NEW POND DRAIN LINES FROM POND KETTLES TO HERRING CREEK.

SURVEY CONTROL COORDINATES				
POINT NO.	RAW/DESC	NORTHING	EASTING	ELEVATION
3366	BM	3647994.67	11864267.86	20.23
3368	BM	3648451.35	11863268.99	28.68
3371	BM	3648977.42	11864094.49	24.68

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	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
OVERALL SITE PLAN AND CONTROL			G007
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
DRAWING NO.			----



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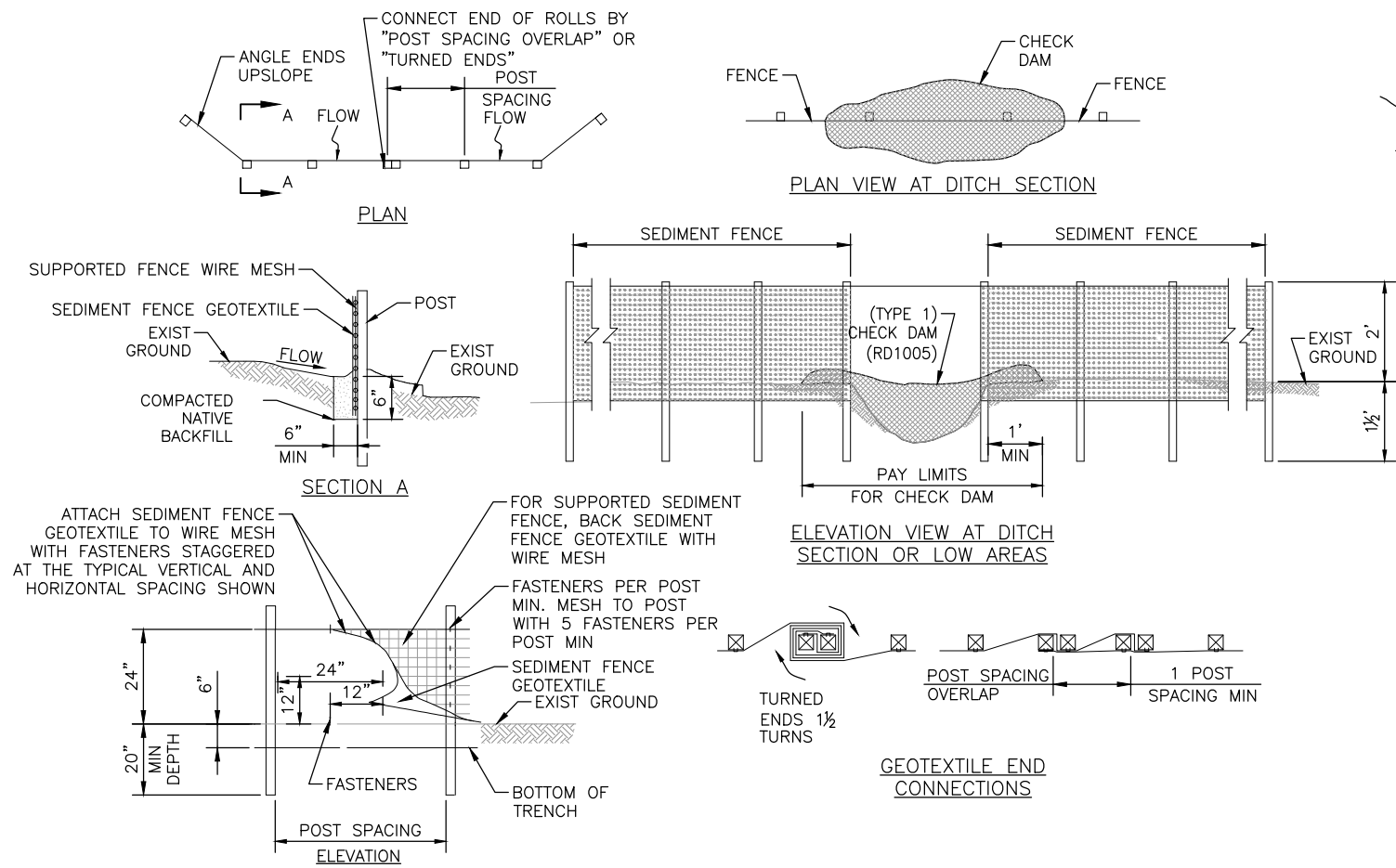


TABLE 1
FENCE SPACING FOR GENERAL APPLICATION

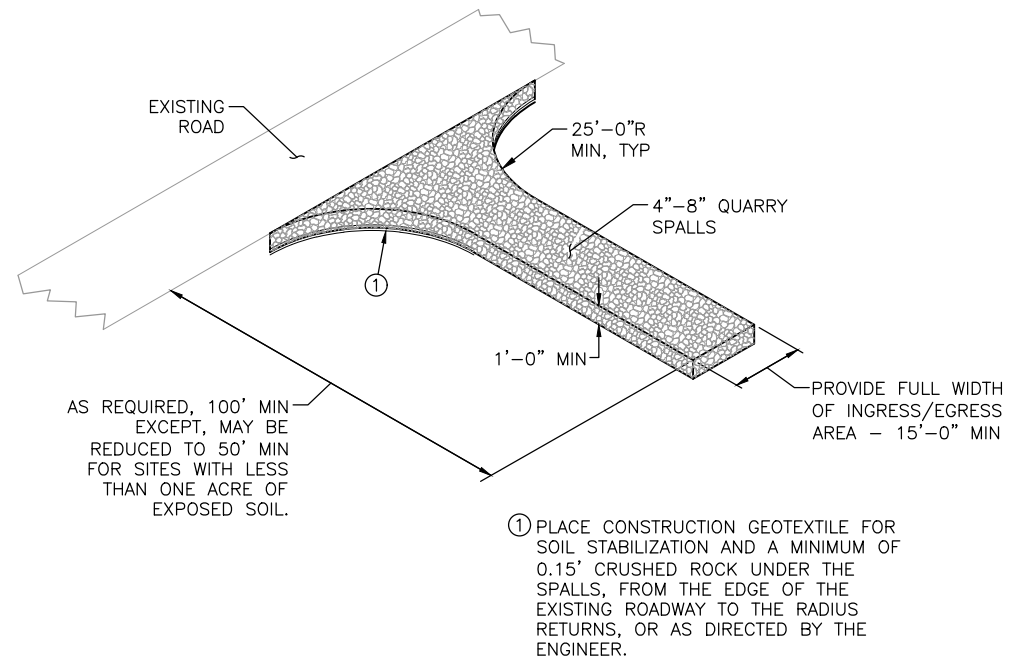
INSTALL PARALLEL ALONG CONTOURS AS FOLLOWS

GRADE	MAX SPACING ON GRADE
GRADE ≤ 0%	300'
10% ≤ GRADE < 15%	150'
15% ≤ GRADE < 20%	100'
20% ≤ GRADE < 30%	50'
30% ≤ GRADE	25'

TABLE 2
POST SPACING

4'	SUPPORTED SEDIMENT FENCE
6'	UNSUPPORTED SEDIMENT FENCE WITH GEOTEXTILE ELONGATION * LESS THAN 50%
4'	UNSUPPORTED SEDIMENT FENCE WITH GEOTEXTILE ELONGATION * MORE THAN 50%

* GEOTEXTILE GRAB ELONGATION VALUE AS DOCUMENTED BY "LEVEL B" MANUFACTURER'S DOCUMENTATION (SEE STANDARD SPECIFICATIONS).

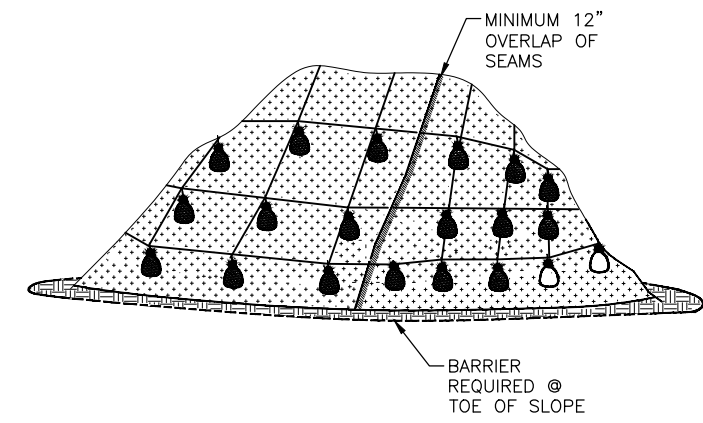


SILT FENCE DETAIL
SCALE: NTS

1
- -

TEMPORARY ENTRANCE DETAIL
SCALE: NTS

2
- -



- NOTES:**
1. MINIMUM 12" OVERLAP OF ALL SEAMS REQUIRED.
 2. BARRIER REQUIRED @ TOE OF STOCK PILE.
 3. COVERING MAINTAINED TIGHTLY IN PLACE BY USING SANDBAGS OR TIRES ON ROPES WITH A MAXIMUM 10' GRID SPACING IN ALL DIRECTIONS.

PLASTIC SHEETING DETAIL
SCALE: NTS

3
- -

CONCEPT DESIGN
NOT FOR CONSTRUCTION

McMillen

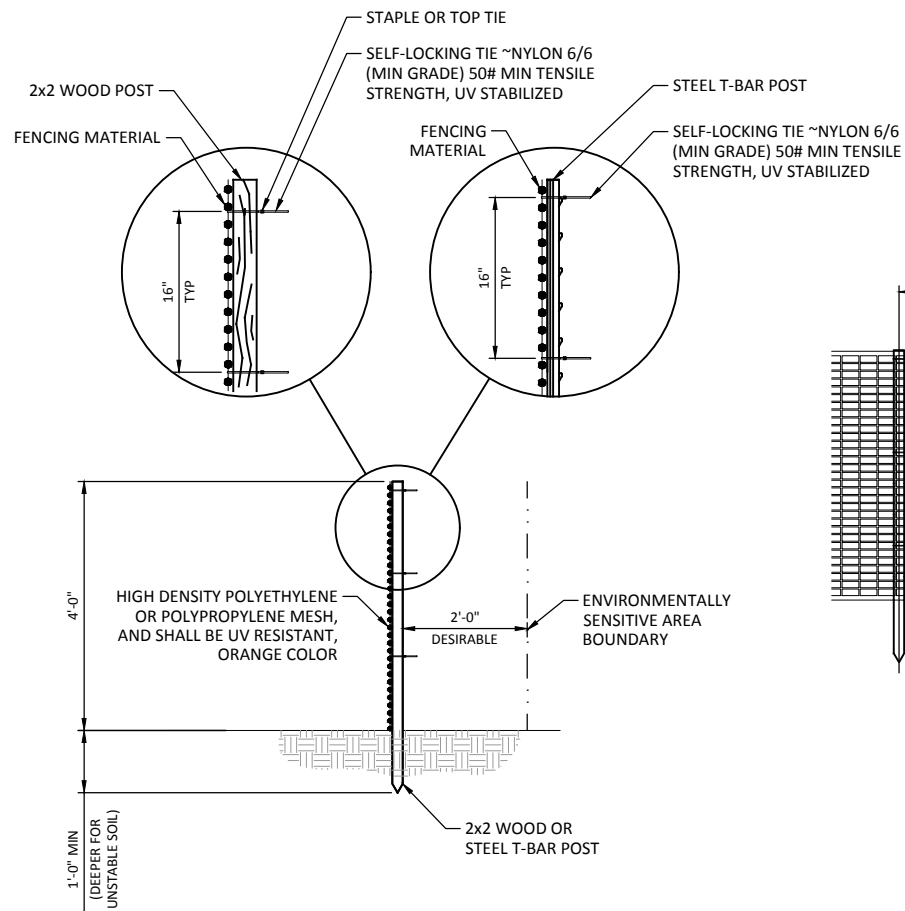
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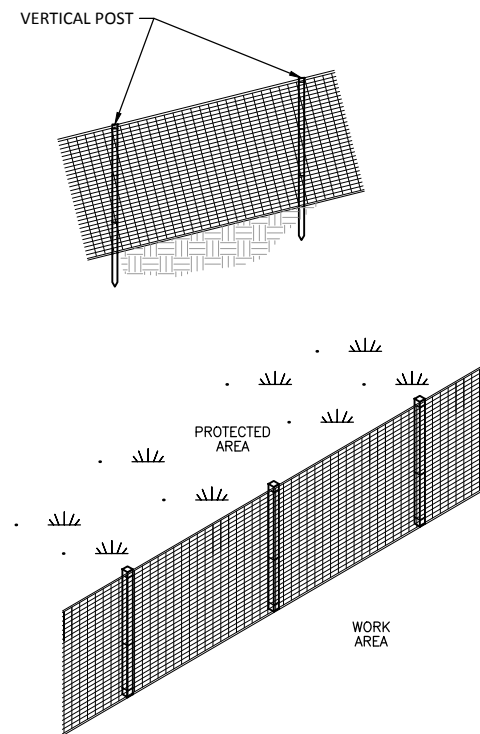
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DRAWN	05/05/23	65% DESIGN SUBMITTAL	JB
CHECKED	10/28/22	35% SCHEMATIC DESIGN	JB
DATE	REV.	DESCRIPTION	BY
HARRISON LAKE NFH - WATER SUPPLY REHABILITATION			
STANDARD EROSION AND SEDIMENT CONTROL DETAILS 1 EC001			
DESIGNED	DATE	CHECKED	DRAWING NO.
RH	07/28/23	JB	



NOTE:

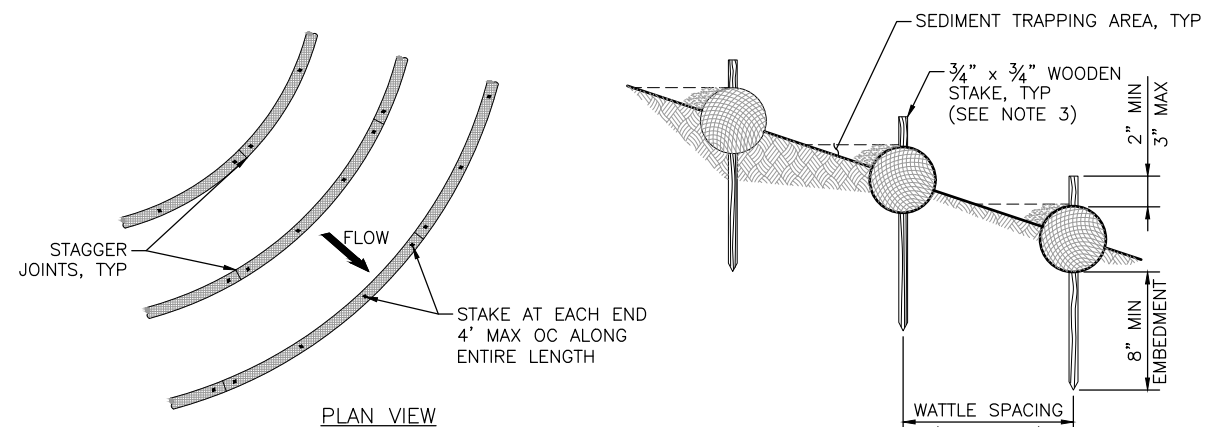
1. POST SHALL HAVE SUFFICIENT STRENGTH AND DURABILITY TO SUPPORT THE FENCE THROUGH THE LIFE OF THE PROJECT.



CONSTRUCTION LIMITS FENCE DETAIL

SCALE: NTS

4



WATTLE SPACING TABLE	
SLOPE	MAXIMUM SPACING
1:1	10 FEET
2:1	20 FEET
3:1	30 FEET
4:1	40 FEET
>4:1	80 FEET

NOTES

1. INSTALL WATTLES ALONG CONTOURS. SEE TABLE FOR SPACING.
2. WATTLES SHALL BE INSPECTED REGULARLY, AND IMMEDIATELY AFTER A RUNOFF PRODUCING RAINFALL, TO ENSURE THEY REMAIN THOROUGHLY ENTRENCHED AND IN CONTACT WITH THE SOIL.
3. LIVE STAKES MAY BE USED FOR PERMANENT INSTALLATIONS.
4. INSTALL WATTLES SNUGLY INTO THE TRENCH. ABUT ADJACENT WATTLES TIGHTLY, END TO END, WITHOUT OVERLAPPING THE ENDS.
5. PILOT HOLES MAY BE DRIVEN THROUGH THE WATTLE AND INTO THE SOIL, WHEN SOIL CONDITIONS REQUIRE.
6. INSTALL AT TOE OF SLOPES. SLOPES GREATER THAN 15' IN LENGTH SHALL HAVE A WATTLE INSTALLED MID SLOPE.

STRAW WATTLE DETAIL

SCALE: NTS

5

CONCEPT DESIGN
NOT FOR CONSTRUCTION

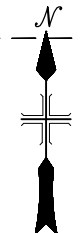
McMillen

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REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
STANDARD EROSION AND SEDIMENT CONTROL DETAILS 2 EC002			
DESIGNED	07/28/23	DRAWING NO.	
RH			
DRAWN			
FH			
CHECKED			
JB			
DATE			
07/28/23			



EROSION AND SEDIMENT CONTROL KEY PLAN



EROSION AND SEDIMENT CONTROL NOTES – BMP MEASURES:

- ALL RUNOFF FROM SITE CONSTRUCTION ACTIVITIES AND FROM RAINFALL EVENTS SHALL BE DETAINED ON SITE AND FILTERED PRIOR TO DISCHARGE. STORMWATER RUNOFF SHALL NOT BE ALLOWED TO LEAVE THE SITE UNTREATED (LADEN W/ SUSPENDED SEDIMENT). IF THIS OCCURS, THE CONTRACTOR WILL BE HELD SOLELY RESPONSIBLE FOR ANY PERMIT VIOLATIONS AND FINES.
- CONTRACTOR SHALL TAKE APPROPRIATE MEASURES TO PREVENT ACCUMULATION OF CONSTRUCTION WASTE AND LITTER ON-SITE.
- CONTRACTOR SHALL INSTALL SILT FENCE AND/OR STRAW WATTLES AS INDICATED IN DETAIL PLANS 1-4 AND IN ANY ADDITIONAL LOCATIONS WHERE MATERIAL COULD LEAVE THE CONSTRUCTION SITE, AT THE CONTRACTOR'S EXPENSE.
- THE SILT FENCE AND/OR STRAW WATTLES SHALL BE INSTALLED PRIOR TO ANY CONSTRUCTION ACTIVITIES.
- CONTRACTOR SHALL HAVE AVAILABLE AT ALL TIMES ADEQUATE SPRINKLER EQUIPMENT TO FACILITATE DUST ABATEMENT AND CONTROL. CONTRACTOR SHALL PROVIDE ALL WATER NECESSARY FOR SPRINKLER OPERATIONS.
- STOCKPILED EXCAVATION MATERIALS SHALL BE PROTECTED FROM WATER AND WIND EROSION BY COVERING AS APPROPRIATE. WHEN EXPOSED FOR MORE THAN 14 DAYS, COVER STOCKPILES WITH IMPERMEABLE TARPS TO PROTECT DISTURBED SOILS AND SLOPES.
- ALL TOP SOIL SHALL BE STRIPPED AND PLACED IN SEPARATE STOCKPILE. AFTER BANK RESTORATION TO EXIST GRADE, TOP SOIL SHALL BE PLACED AND RESEED.
- CONTRACTOR SHALL HAVE ON-SITE AT ALL TIMES SPILL PREVENTION AND CONTROL MEASURES.
- ENSURE ALL EQUIPMENT IS CLEAN AND FREE OF OIL/FUEL LEAKS, DIRT, PLANTS AND ANIMALS OR FRAGMENTS OF PLANTS, AQUATIC INVASIVE SPECIES, AND OTHER VEGETATIVE MATTER. EQUIPMENT FOR WORK INSIDE OF THE ORDINARY HIGH WATER MARK SHALL UTILIZE FOOD-GRADE HYDRAULIC FLUID.

EROSION AND SEDIMENT CONTROL NOTES – GRADING AND FINAL STABILIZATION:

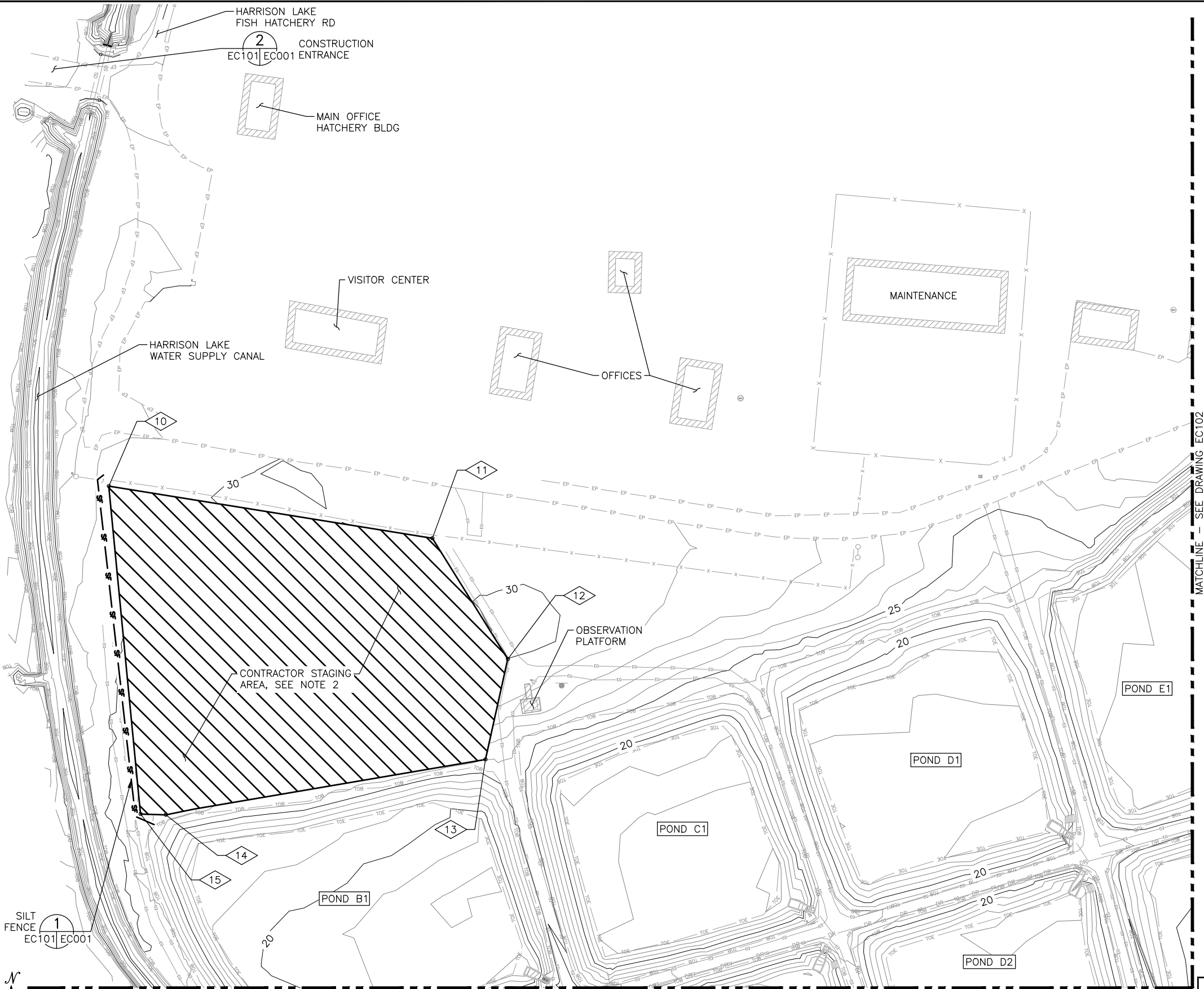
- CLEARING, GRUBBING, AND GROUND DISTURBING ACTIVITIES SHALL BE CONFINED TO WITHIN THE CLEARING LIMITS AND SHALL MEET THE REQUIREMENTS OF SPECIFICATION 31 11 00. NO GRADING OR CONSTRUCTION ACTIVITIES SHALL OCCUR OUTSIDE OF THE PROPOSED IMPROVEMENTS SHOWN ON THE CONSTRUCTION PLANS FOR THIS PROJECT. PRESERVE EXIST VEGETATION BEYOND DISTURBED AREA – UTILIZE AS NATURAL BUFFER STRIPS.
- DURING CONSTRUCTION, PROVIDE POSITIVE DRAINAGE AWAY FROM FACILITIES.
- CONTRACTOR SHALL REMOVE ALL TEMPORARY EROSION AND SEDIMENT CONTROL FACILITIES, FENCING, AND STAGING AREA MATERIALS WHEN CONSTRUCTION IS COMPLETE. NO CONSTRUCTION DEBRIS, DEMOLITION MATERIALS, OR EXCESS EQUIPMENT SHALL BE LEFT ON SITE.
- CONTRACTOR SHALL REGRADE DISTURBED SLOPES TO NEAR EXIST CONDITION AS APPROVED BY THE OWNER.
- ESTABLISH A TEMPORARY VEGETATIVE COVER ON ALL DISTURBED AREAS AS SOON AS PRACTICAL AFTER THE LAST GROUND DISTURBING ACTIVITIES IN THE AREA. CONTRACTOR SHALL RESEED ALL DISTURBED AREAS WITH NATIVE VEGETATION.

EROSION AND SEDIMENT CONTROL NOTES – GENERAL:

- THE CONTRACTOR SHALL SUBMIT AN EROSION AND SEDIMENT CONTROL PLAN FOR WORK DURING CONSTRUCTION THAT MEETS ALL FEDERAL, STATE, AND LOCAL REQUIREMENTS.
 - THE CONTRACTOR IS RESPONSIBLE FOR IMPLEMENTATION AND MAINTENANCE OF EROSION AND SEDIMENT CONTROL MEASURES (MULCHING OF STRAW, SAND DIVERSION DITCHES, ETC.) DICTATED BY FIELD CONDITIONS TO PREVENT EROSION OR THE INTRODUCTION OF DIRT, MUD, OR DEBRIS TO EXIST PUBLIC OR PRIVATE ROADWAY, ONTO ADJACENT PROPERTIES, INTO THE CREEK OR CANAL DURING ANY PHASE OF CONSTRUCTION OPERATIONS. SPECIAL ATTENTION SHALL BE GIVEN TO ADDITIONAL EROSION AND SEDIMENT CONTROL MEASURES NOTED BELOW.
 - THE GENERAL EROSION AND SEDIMENT CONTROL PLAN ON THE EC DRAWINGS ARE PROVIDED TO AID THE CONTRACTOR IN DEVELOPING THE EROSION AND SEDIMENT CONTROL PLAN ACCORDING TO CONTRACTOR SCHEDULE AND PHASING OF THE PROJECT.
 - EROSION CONTROL DETAILS ARE FOR INFORMATION ONLY TO AID THE CONTRACTOR. THE FINAL LOCATIONS AND DETAIL SHALL BE SHOWN ON THE CONTRACTOR'S PREPARED STORMWATER POLLUTION PREVENTION PLAN (SWPPP) DOCUMENT.
 - CONTRACTOR IS RESPONSIBLE TO PROVIDE ALL NECESSARY EROSION CONTROL MEASURES FOR THE DURATION OF THE PROJECT. MAINTENANCE OF BOTH TEMPORARY AND PERMANENT EROSION CONTROL MEASURES SHALL BE CONSIDERED INCIDENTAL.
 - ALL BMP REQUIRED MATERIALS SHALL MEET OR EXCEED STATE OF VIRGINIA STORMWATER REQUIREMENTS.
 - CONTRACTOR SHALL DEVELOP A SPILL PREVENTION, CONTAINMENT, AND RESPONSE PLAN THAT WILL BE ATTACHED TO THE SWPPP.

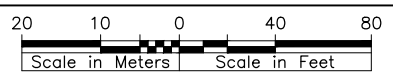
VERIFY SCALE
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ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET

	07/28/23	95% DESIGN SUBMITTAL	RH
	05/05/23	65% DESIGN SUBMITTAL	JB
	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
EROSION AND SEDIMENT CONTROL KEY PLAN			EC100
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23
			DRAWING NO. ----



EROSION AND SEDIMENT CONTROL PLAN 1

1
EC100|EC101



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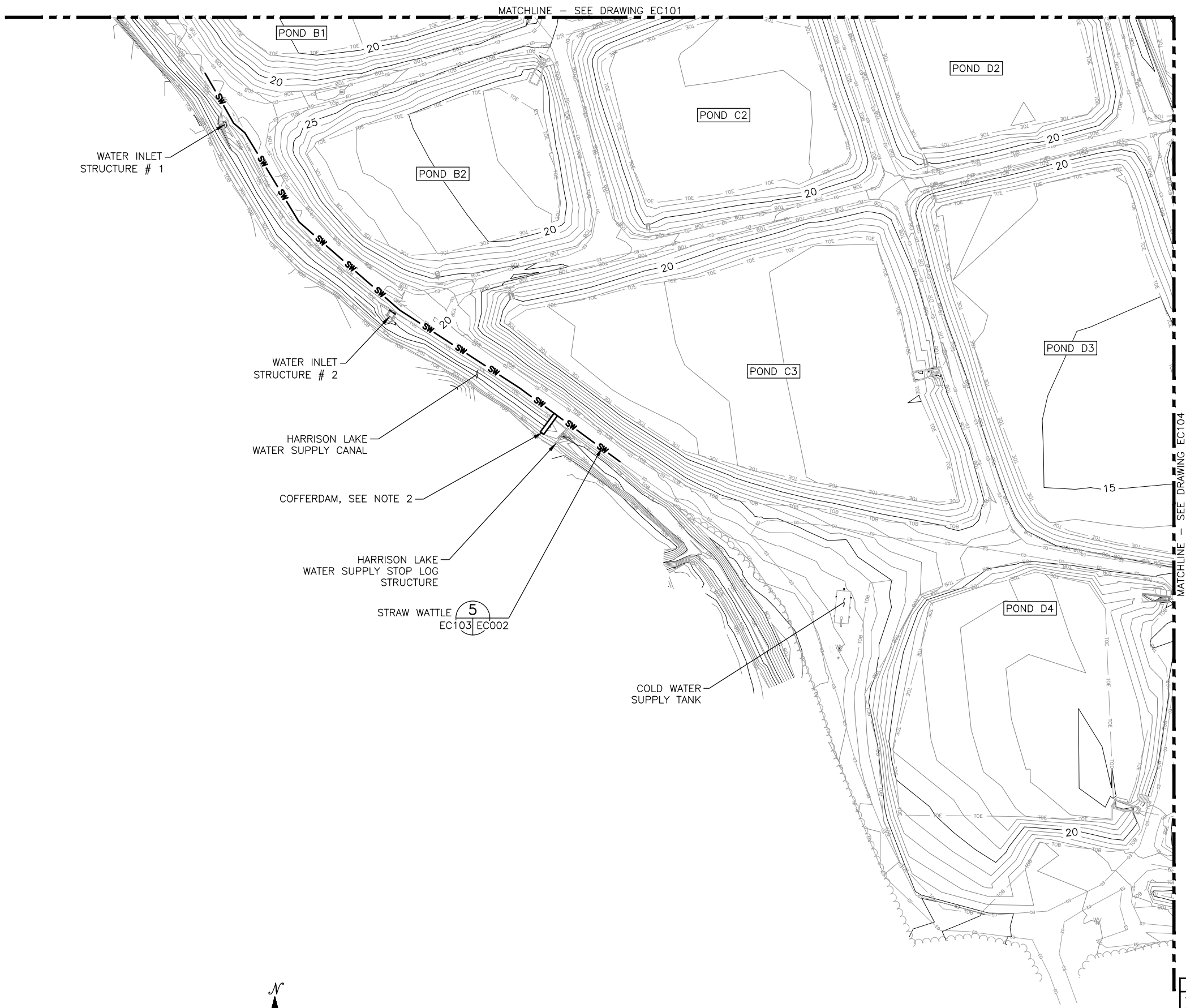
SHEET NOTES:

1. CONTRACTOR SHALL USE CONSTRUCTION FENCE, AS REQUIRED, TO PRECLUDE THE PUBLIC FROM THE CONSTRUCTION AREA DURING DEMO OF THE EXISTING AC PIPE.
2. CONTRACTOR STAGING AREA SHALL BE FOR CONTRACTOR'S EMPLOYEE PARKING, CONTRACTOR'S TRAILERS, AND ONSITE STORAGE OF MATERIALS. LOCATION OF STAGING AREA AND ACCESS TO SITE MUST BE APPROVED BY OWNER.
3. CONTRACTOR SHALL INSTALL ALL ESC MEASURES PRIOR TO INITIATING FIELD CONSTRUCTION ACTIVITIES.

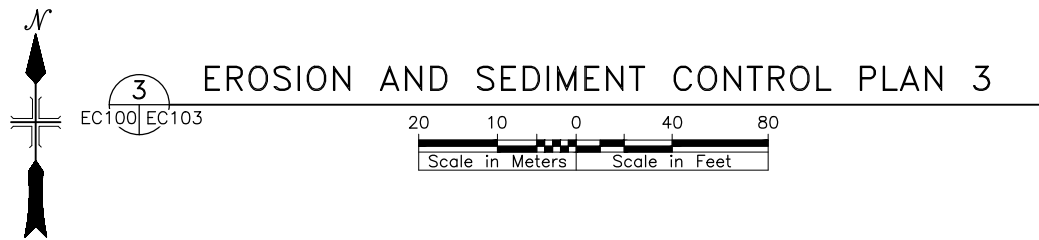
CONTRACTOR STAGING AREA COORDINATE POINT TABLE		
POINT #	NORTHING	EASTING
10	3648903.96	11863167.34
11	3648865.39	11863406.47
12	3648776.14	11863462.16
13	3648701.99	11863445.65
14	3648661.04	11863209.47
15	3648661.61	11863190.81

CONCEPT DESIGN
NOT FOR CONSTRUCTION

	07/28/23	95% DESIGN SUBMITTAL		RH
	05/05/23	65% DESIGN SUBMITTAL		JB
	10/28/22	35% SCHEMATIC DESIGN		JB
REV.	DATE	DESCRIPTION		BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION				
EROSION AND SEDIMENT CONTROL PLAN 1				EC101
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23	DRAWING NO. _____



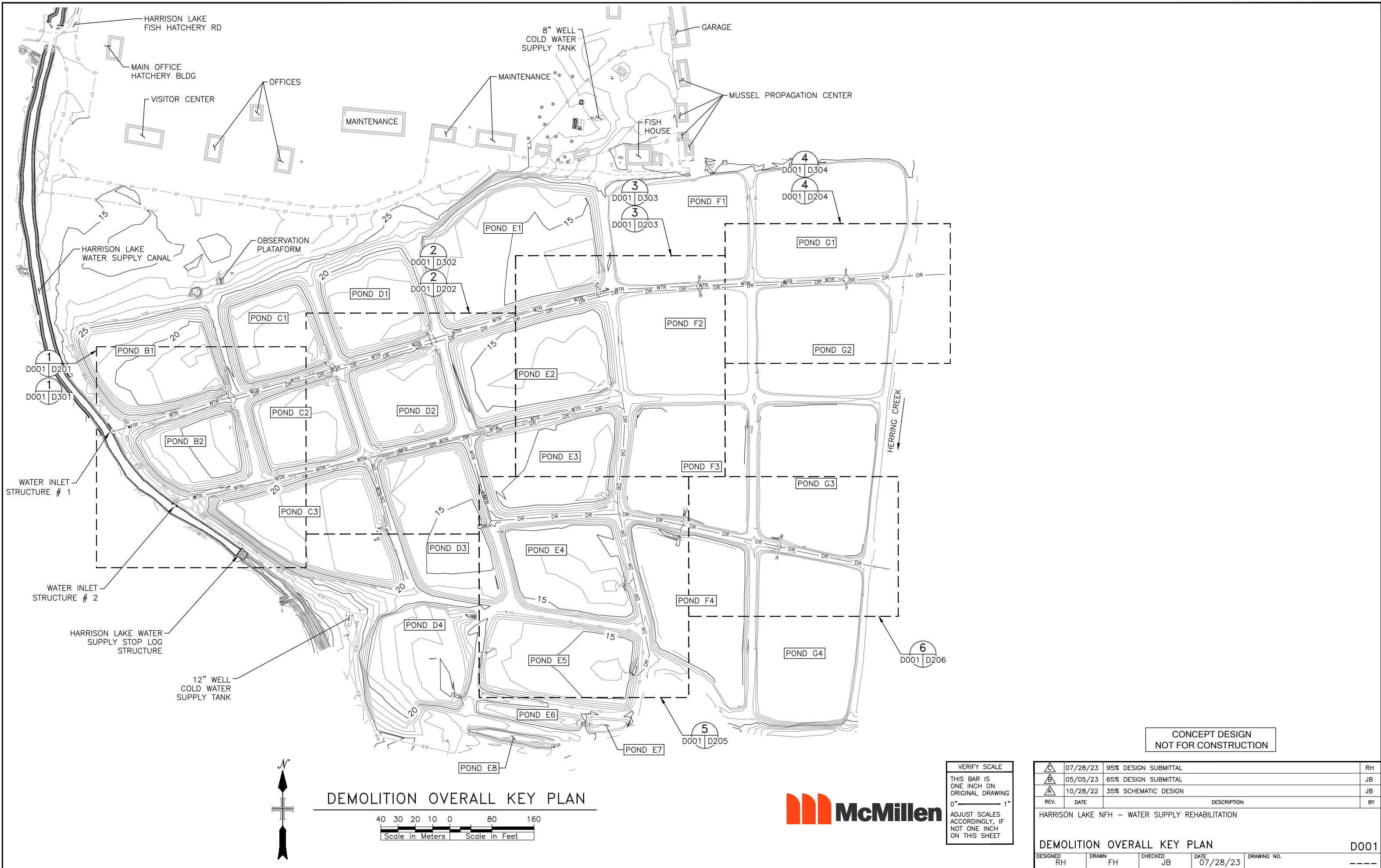
- SHEET NOTES:**
1. CONTRACTOR SHALL USE CONSTRUCTION FENCE, AS REQUIRED, TO PRECLUDE THE PUBLIC FROM THE CONSTRUCTION AREA DURING DEMO OF THE EXISTING AC PIPE.
 2. CONTRACTOR SHALL INSTALL COFFER DAM TO MAINTAIN WATER SUPPLY TO THE HATCHERY DURING CONSTRUCTION OF THE STOP LOG STRUCTURE. CONTRACTOR SHALL SUBMIT THE COFFERDAM PLAN FOR APPROVAL.

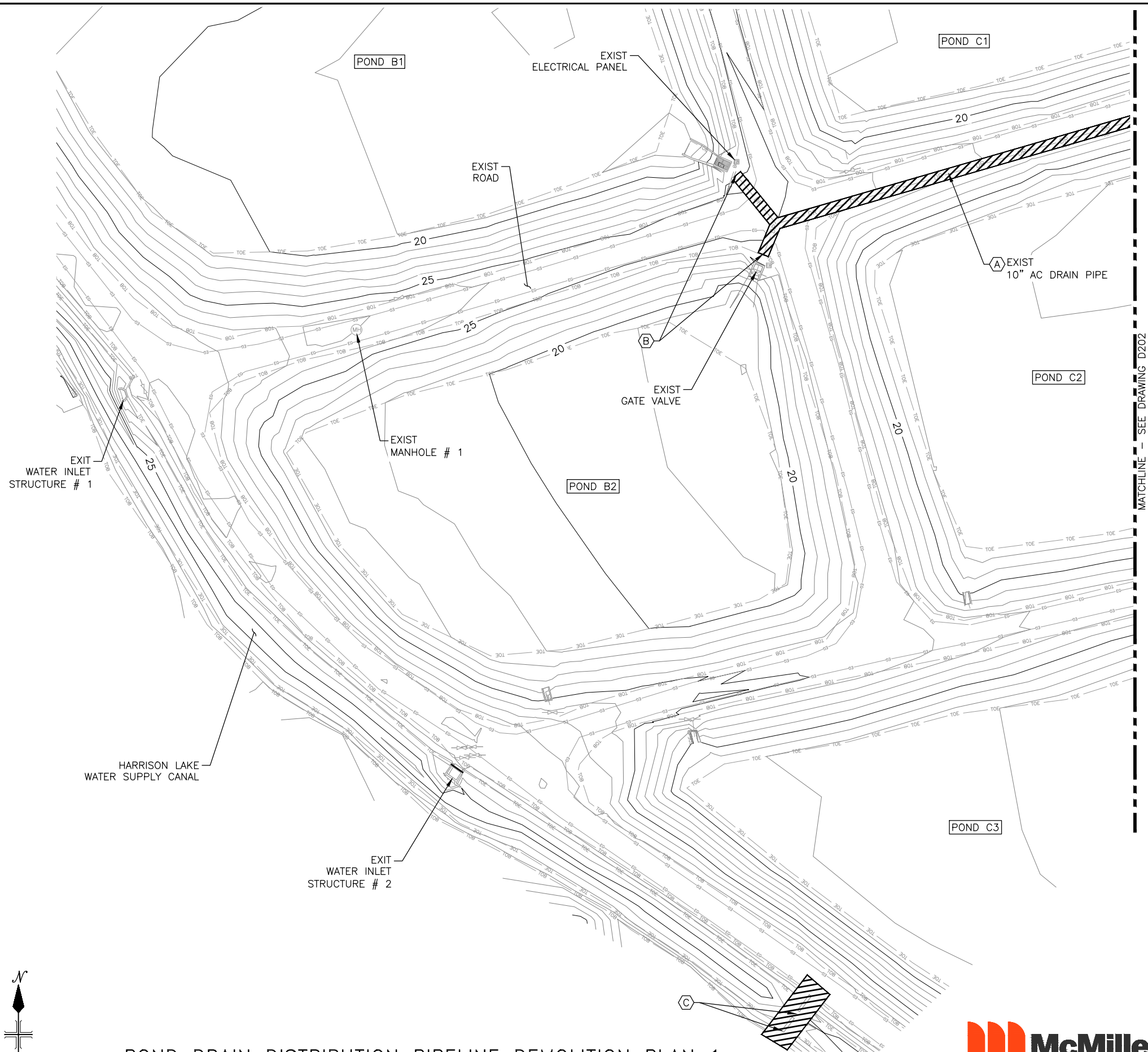


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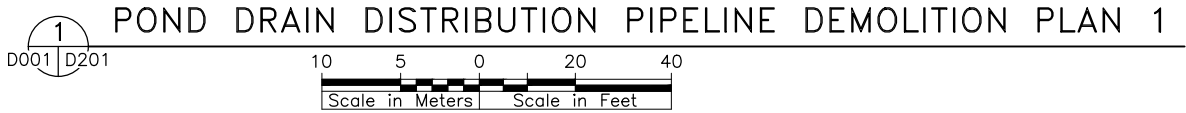
DESIGNED	DATE	CHECKED	DATE	DESCRIPTION	BY
RH	07/28/23	JB	05/05/23	95% DESIGN SUBMITTAL	RH
			10/28/22	65% DESIGN SUBMITTAL	JB
				35% SCHEMATIC DESIGN	JB
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION					
EROSION AND SEDIMENT CONTROL PLAN 3					EC103
DESIGNED	DATE	CHECKED	DATE	DESCRIPTION	BY
RH	07/28/23	JB	07/28/23	EROSION AND SEDIMENT CONTROL PLAN 3	JB





- SHEET NOTES:**
1. CONTRACTOR SHALL DEMOLISH AND DISPOSE OF EXISTING 10-INCH DIAMETER DRAIN PIPE. DISPOSE OF MATERIALS OFFSITE AT APPROVED DISPOSAL LOCATION.
 2. CONTRACTOR TO FIELD VERIFY DRAIN PIPE MATERIAL. IF A.C. CONTRACTOR SHALL TAKE NECESSARY EFFORTS TO SAFELY DISPOSE OF THE PIPE.

- SHEET KEY NOTES:**
- A. REMOVE EXISTING AC 10-INCH DRAIN PIPE .
 - B. REMOVE EXISTING DRAIN LINE TO EXISTING COUPLING.
 - C. DEMOLISH EXISTING STOP LOG STRUCTURE.



VERIFY SCALE

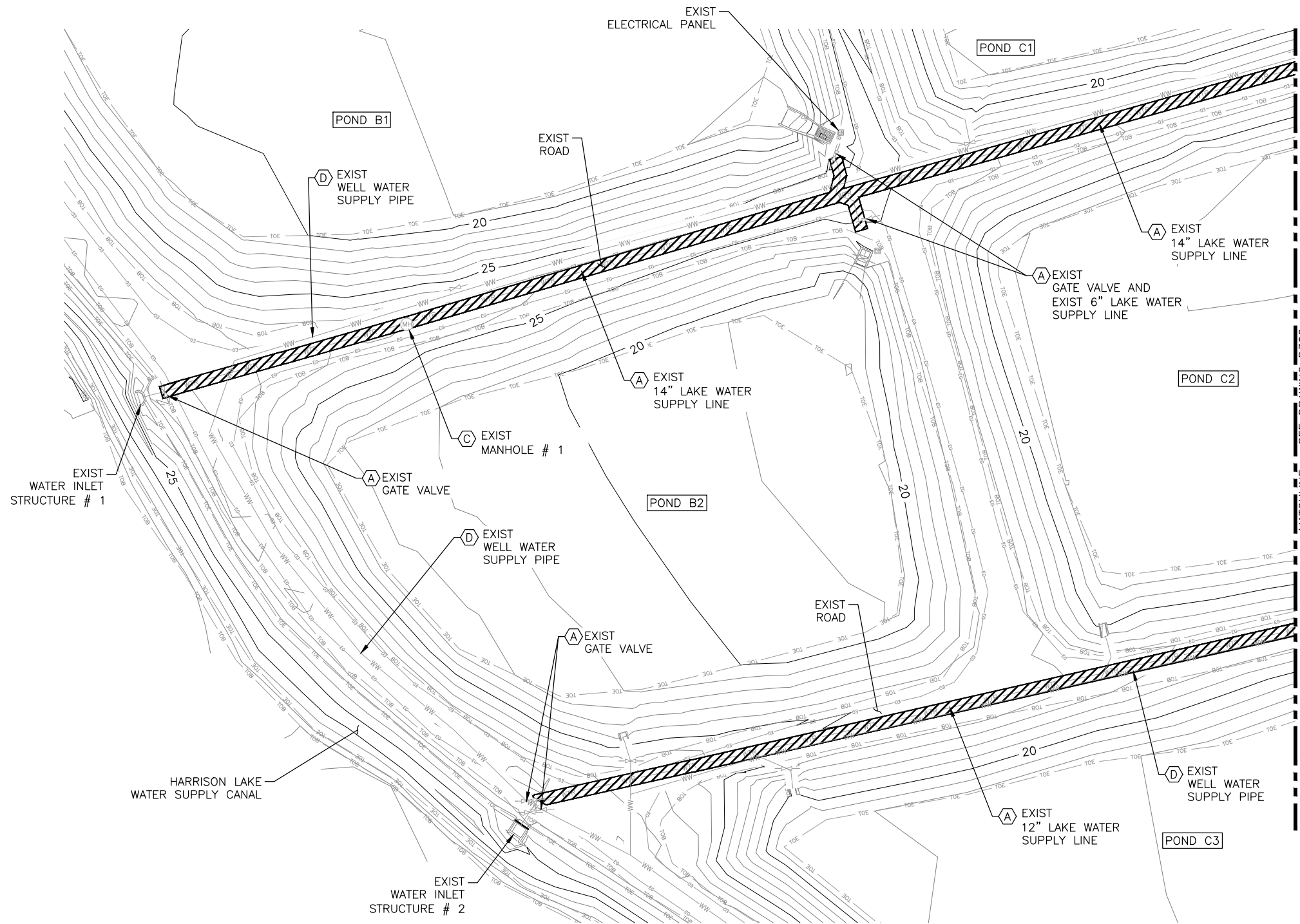
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	10/28/22	35% SCHEMATIC DESIGN				JB
REV.	DATE	DESCRIPTION				BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION						
POND DRAIN DISTRIBUTION PIPELINE DEMOLITION PLAN 1						D201
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23	DRAWING NO.		



SHEET NOTES:

1. CONTRACTOR SHALL DEMOLISH AND DISPOSE OF EXISTING 6-INCH DIAMETER WATER DISTRIBUTION PIPE. DISPOSE OF MATERIALS OFFSITE AT APPROVED DISPOSAL LOCATION.
2. CONTRACTOR TO FIELD VERIFY WATER DISTRIBUTION PIPE MATERIAL OF WROUGHT IRON.

SHEET KEY NOTES:

- A. REMOVE EXISTING WROUGHT IRON LAKE WATER SUPPLY PIPE TO EXISTING GATE VALVE. RETAIN AND PROTECT EXISTING GATE VALVE UNLESS OTHERWISE INDICATED.
- B. REMOVE AND REPLACE GATE VALVE IF NOT REPLACED IN 2021.
- C. DEMO EXISTING MANHOLE.
- D. RETAIN AND PROTECT EXISTING WELL WATER SUPPLY PIPE.



1
D001 D301

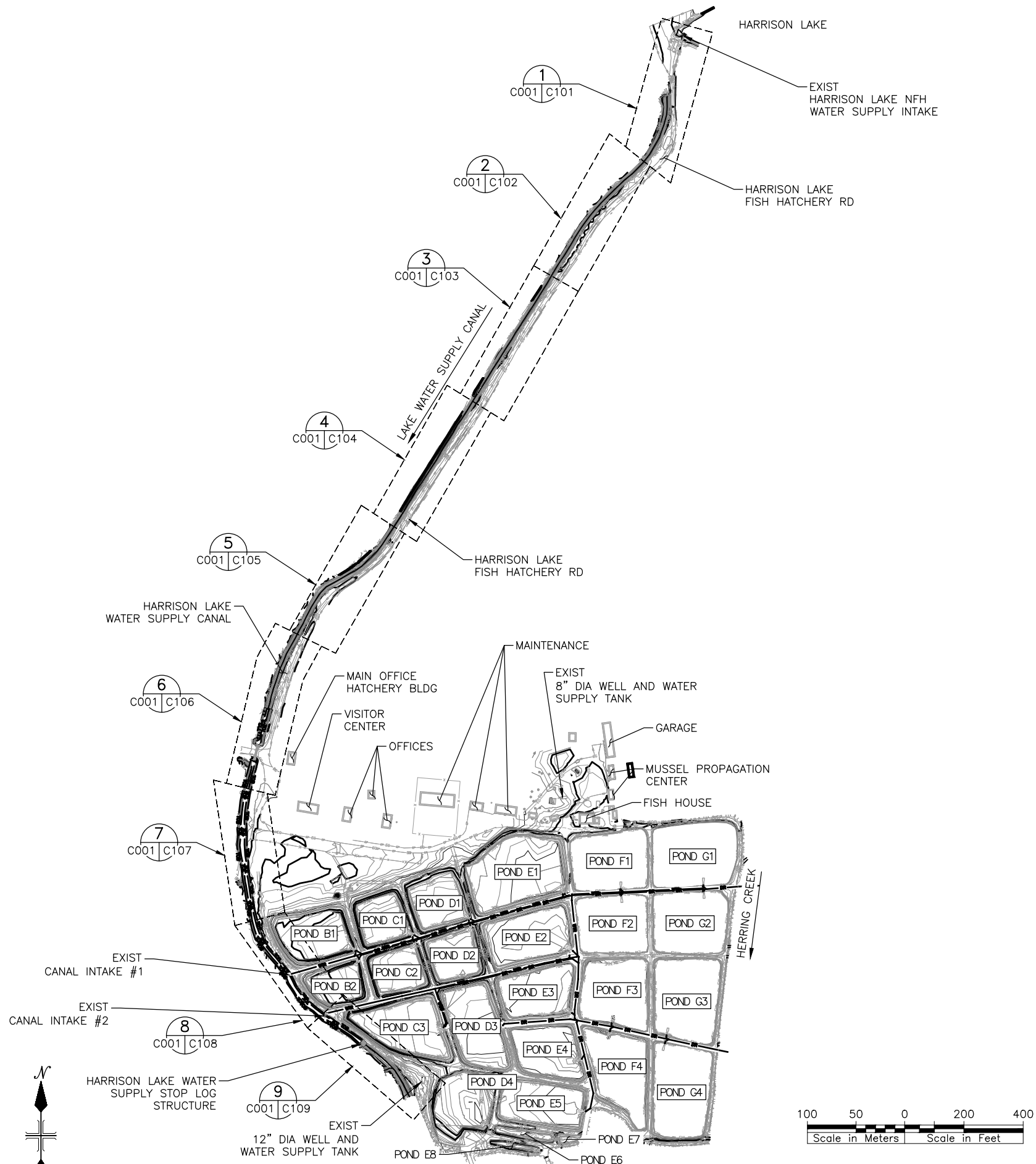
LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 1



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10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION		
LAKE WATER DISTRIBUTION PIPELINE DEMOLITION PLAN 1		
DESIGNED RH	DRAWN FH	CHECKED JB
DATE 07/28/23	DRAWING NO. D301	



LAKE WATER SUPPLY CANAL OVERALL SITE KEY PLAN

CONCEPT DESIGN
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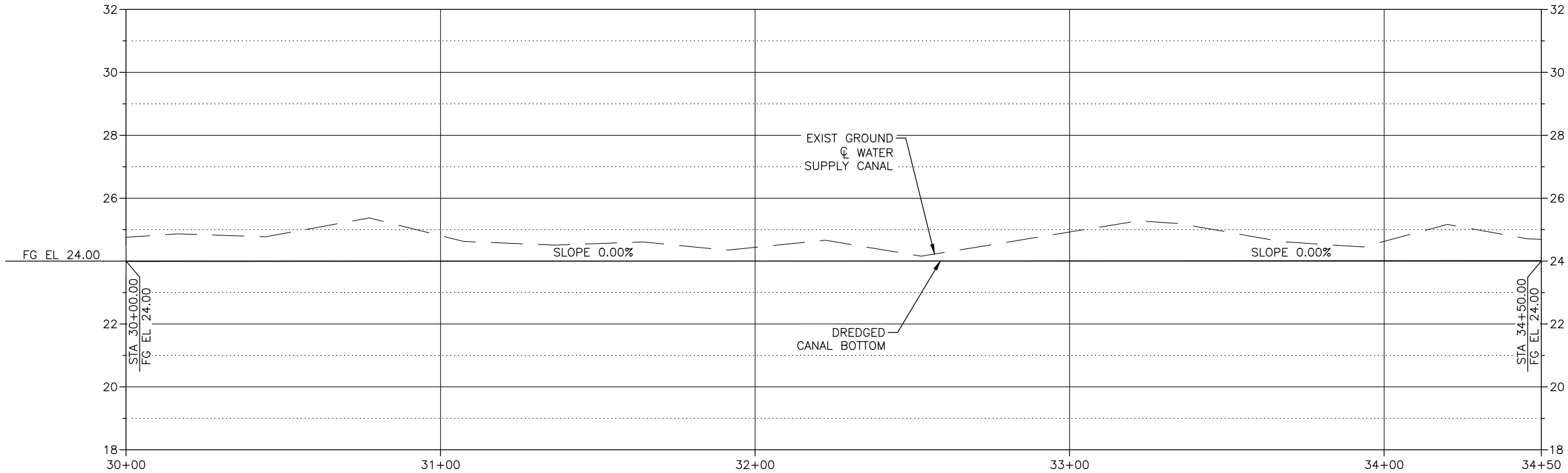
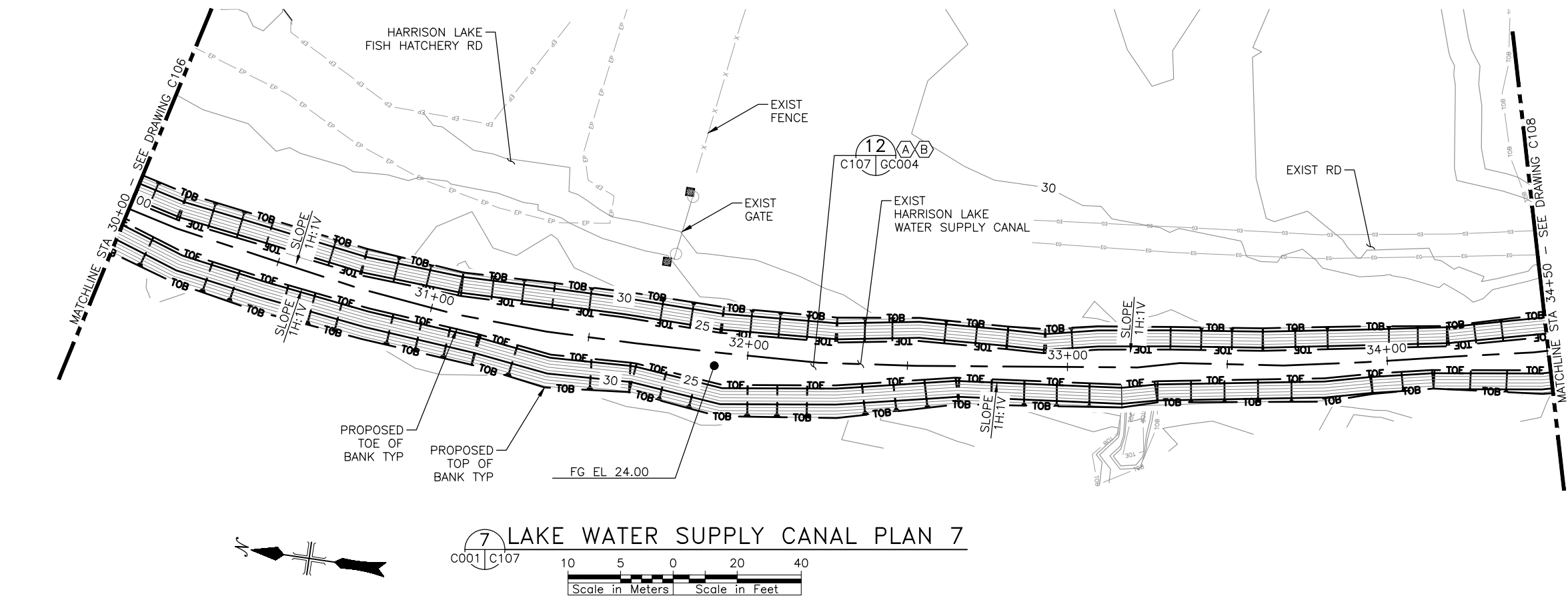
DESIGNED	DATE	REV.	DESCRIPTION	BY
RH	07/28/23	A	95% DESIGN SUBMITTAL	RH
FB	05/05/23	B	65% DESIGN SUBMITTAL	JB
FB	10/28/22	A	35% SCHEMATIC DESIGN	JB
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION				
LAKE WATER SUPPLY CANAL OVERALL SITE KEY PLAN C001				
DESIGNED	DRAWN	CHECKED	DATE	DRAWING NO.
RH	FH	JB	07/28/23	----

SHEET NOTES:

1. CONTRACTOR SHALL REPAIR EXISTING CANAL BANKS AND DREDGE BOTTOM OF CHANNEL.

SHEET KEY NOTES:

- A. WHERE BANK IS BELOW EL 29.00, REBUILD CANAL TOP OF BANK UP TO EL. 29.00 FT.
- B. DREDGE CHANNEL BOTTOM TO ELEVATION 24.00 TO PROPOSED STOP LOG STRUCTURE.



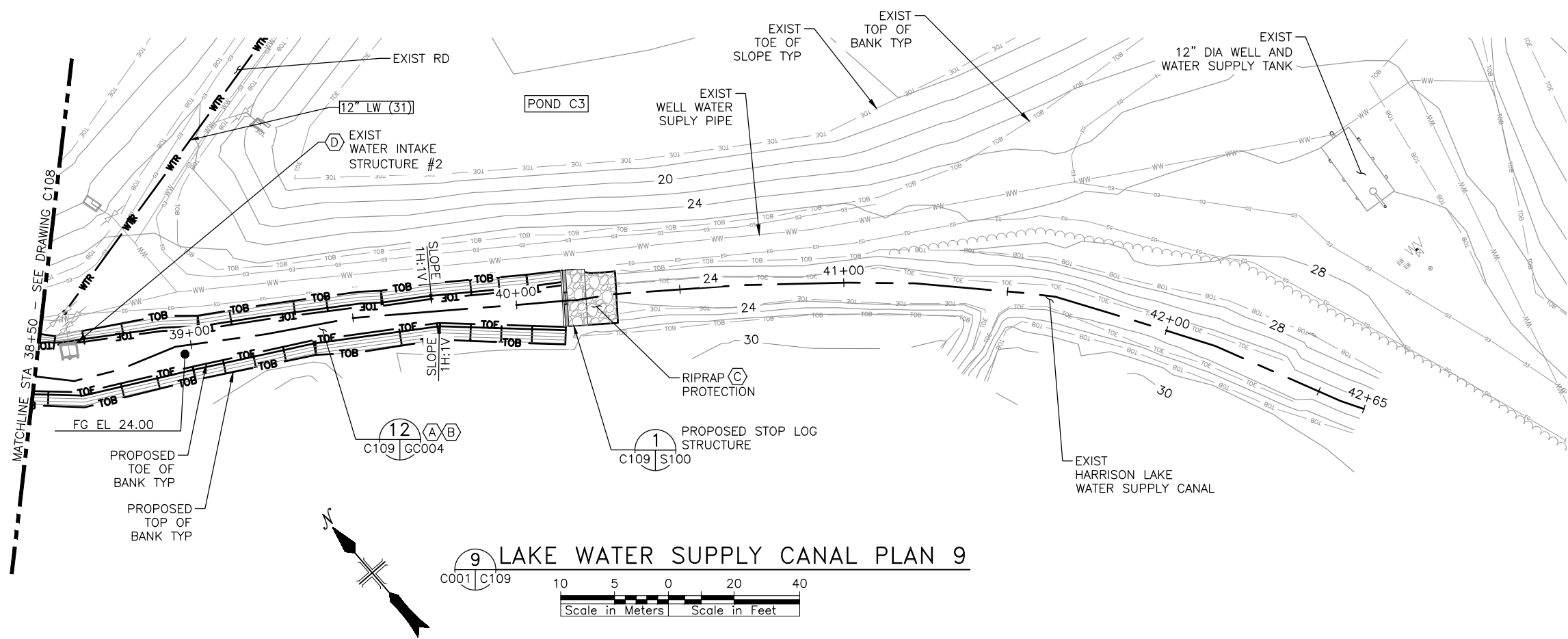
CONCEPT DESIGN
NOT FOR CONSTRUCTION

LAKE WATER SUPPLY CANAL PROFILE 7

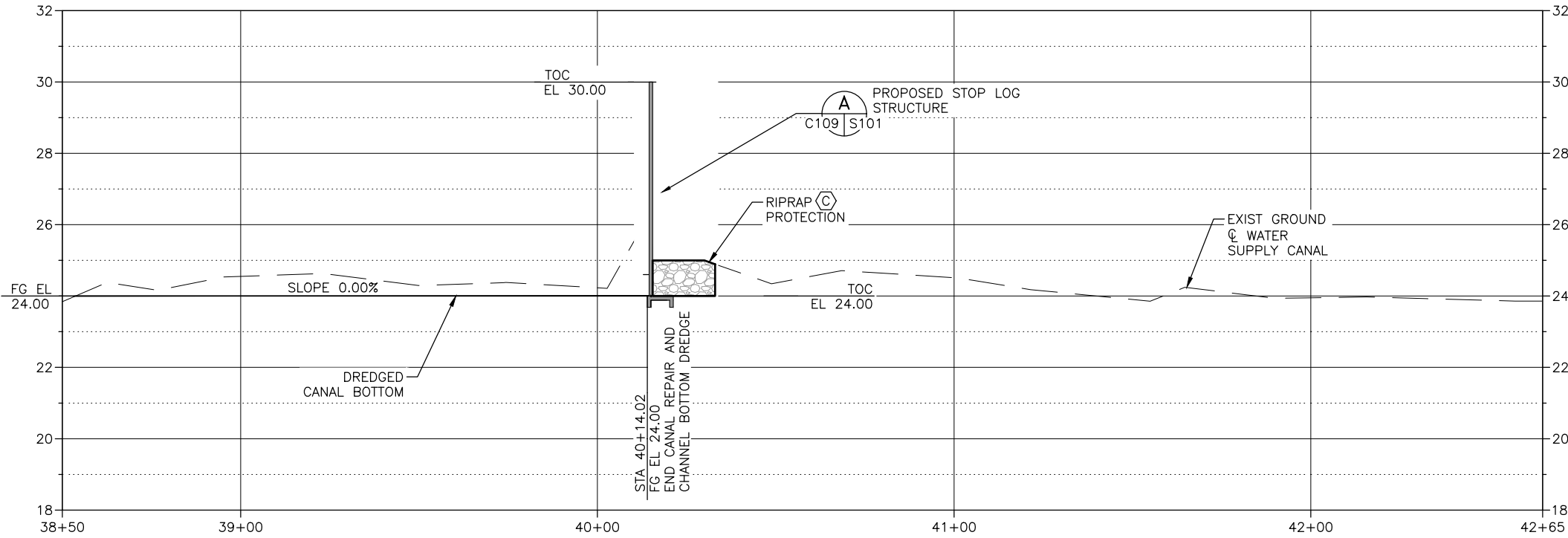


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05/05/23	65% DESIGN SUBMITTAL	JB
10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION
HARRISON LAKE NFH - WATER SUPPLY REHABILITATION		
LAKE WATER SUPPLY CANAL PLAN AND PROFILE 7 C107		
DESIGNED RH	DRAWN FH	CHECKED JB
DATE 07/28/23	DRAWING NO. ----	



- SHEET NOTES:
- CONTRACTOR SHALL REPAIR EXISTING CANAL BANKS AND DREDGE BOTTOM OF CHANNEL.
- SHEET KEY NOTES:
- WHERE BANK IS BELOW EL. 29.0, REBUILD CANAL TOP OF BANK UP TO EL. 29.00 FT.
 - DREDGE CHANNEL BOTTOM TO ELEVATION 24.00 SOUTH OF EXISTING 72" CULVERT. TO PROPOSED STOP LOG STRUCTURE.
 - PLACE CLASS I RIPRAP BANK PROTECTION DOWNSTREAM OF STOP LOG STRUCTURE. SEE SPECIFICATIONS FOR SIZING.
 - RETAIN AND PROTECT EXISTING INTAKE STRUCTURE.



LAKE WATER SUPPLY CANAL PROFILE 9



CONCEPT DESIGN
NOT FOR CONSTRUCTION



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	10/28/22	35% SCHEMATIC DESIGN					JB
REV.	DATE	DESCRIPTION					BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION							
LAKE WATER SUPPLY CANAL PLAN AND PROFILE 9 C109							
DESIGNED RH	DRAWN FH	CHECKED JB	DATE 07/28/23	DRAWING NO. -----			

GENERAL STRUCTURAL NOTES:
THE FOLLOWING NOTES ARE GENERAL AND APPLY TO THE ENTIRE PROJECT, UNLESS SPECIFICALLY NOTED OTHERWISE (UNO)

1) GENERAL:

A. CONSTRUCTION DOCUMENTS:

1. THE CONTRACTOR SHALL REVIEW THE APPROVED CONTRACT DOCUMENTS AND NOTIFY THE ENGINEER OF ANY ERRORS OR DISCREPANCIES PRIOR TO THE START OF CONSTRUCTION.
2. THE CONTRACTOR SHALL FURNISH AND INSTALL EVERYTHING REQUIRED TO PROVIDE A COMPLETE STRUCTURE AS SHOWN HEREIN. IF THERE IS AN OMISSION ON THE PLANS, SUCH OMISSION SHALL NOT BE CONSTRUED TO MEAN THAT THE CONTRACTOR IS NOT REQUIRED TO FURNISH OR PROVIDE EVERYTHING THAT IS NECESSARY TO COMPLETE THE PROJECT TO THE MINIMUM REQUIREMENTS OF THE IBC AND ALL OTHER SPECIFICATIONS, CODES AND STANDARDS NOTED ON THE APPROVED CONTRACT DOCUMENTS.
3. THE CONTRACTOR SHALL NOTIFY THE GOVERNMENT IMMEDIATELY IF ANY UNIDENTIFIED EXISTING UNDERGROUND UTILITIES ARE DISCOVERED. THE ENGINEER IS NOT RESPONSIBLE FOR THE LOCATIONS OF EXISTING UNDERGROUND UTILITIES WHETHER OR NOT SHOWN ON THE DRAWINGS.
4. THE STRUCTURAL CONTRACT DOCUMENTS REPRESENT THE FINISHED STRUCTURE. THEY DO NOT INDICATE THE METHOD OF CONSTRUCTION. THE CONTRACTOR SHALL PROVIDE ALL MEASURES NECESSARY TO PROTECT THE STRUCTURE DURING CONSTRUCTION. SUCH MEASURES SHALL INCLUDE, BUT ARE NOT LIMITED TO, BRACING AND/OR SHORING FOR LOADS DUE TO CONSTRUCTION EQUIPMENT, ETC. CONTRACTOR AT HIS/HER OWN EXPENSE SHALL ENGAGE PROPERLY QUALIFIED PERSONS TO DESIGN BRACING, SHORING, ETC. OBSERVATION VISITS TO THE SITE BY THE ENGINEER SHALL NOT INCLUDE OBSERVATION OF THE ABOVE NOTED ITEMS.
5. UNDER NO CIRCUMSTANCES CAN STRUCTURAL COMPONENTS BE SUBSTITUTED, OMITTED, OR ALTERED FROM THE APPROVED SET OF CONSTRUCTION DOCUMENTS WITHOUT WRITTEN APPROVAL FROM THE ENGINEER.

B. DIMENSIONS AND NOTATIONS:

1. WRITTEN DIMENSIONS SHALL TAKE PRECEDENCE OVER SCALED DIMENSIONS. DO NOT SCALE DRAWINGS.
2. ABBREVIATIONS USED ON THE APPROVED CONTRACT DOCUMENTS SHALL BE CONSIDERED TYPICAL ABBREVIATIONS FOR THE INDUSTRY. THE CONTRACTOR SHALL BE RESPONSIBLE TO NOTIFY THE ENGINEER AND USFWS IMMEDIATELY OF ANY ABBREVIATIONS THAT ARE UNKNOWN TO THE CONTRACTOR.

C. SHOP DRAWINGS:

1. SHOP DRAWINGS, AS REQUIRED PER THESE STRUCTURAL NOTES, SHALL BE SUBMITTED TO THE USFWS IN A TIMELY FASHION PRIOR TO FABRICATION TO ALLOW FOR PROPER REVIEW AS REQUIRED PER SECTION 107.3.4.1 OF THE IBC (DEFERRED SUBMITTALS).
2. SHOP DRAWING ITEMS SHALL NOT BE INSTALLED UNTIL THE CONSTRUCTION DOCUMENTS HAVE BEEN APPROVED BY THE BUILDING OFFICIAL AND SHOP DRAWINGS HAVE BEEN APPROVED BY THE USFWS PER SECTION 107.3.4 OF THE IBC.
3. DURING SHOP DRAWING REVIEW, DIMENSIONS AND QUANTITIES ARE NOT REVIEWED BY THE USFWS AND MUST BE VERIFIED BY THE CONTRACTOR. THE CONTRACTOR SHALL REVIEW AND STAMP SHOP DRAWINGS PRIOR TO REVIEW BY USFWS.

D. SPECIAL INSPECTION:

1. THE GOVERNMENT SHALL EMPLOY A SPECIAL INSPECTION SERVICE AS REQUIRED PER THESE STRUCTURAL NOTES. REQUIREMENT FOR SPECIAL INSPECTIONS IS THE DECISION OF THE GOVERNING BUILDING OFFICIAL AND IS NOT THE DECISION OF THE ENGINEER. THE LIST OF SPECIAL INSPECTION TASKS IS PROVIDED ONLY IN CASE SPECIAL INSPECTIONS ARE REQUIRED AND DO NOT CONSTITUTE A DIRECTION BY THE ENGINEER TO ANY PARTY THAT SPECIAL INSPECTIONS ARE REQUIRED.

E. TYPICAL NOTES AND DETAILS:

1. SPECIFIC NOTES AND DETAILS SHALL TAKE PRECEDENCE OVER STANDARD TYPICAL NOTES AND DETAILS.
2. STANDARD TYPICAL NOTES AND DETAILS ARE TO BE USED WHEN REFERRED TO OR WHEN NO OTHER MORE RESTRICTIVE OR DIFFERENT DETAILS ARE SHOWN ON THE DRAWINGS.
3. WORK NOT PARTICULARLY SHOWN OR SPECIFIED SHALL BE THE SAME AS SIMILAR PARTS THAT ARE SHOWN OR SPECIFIED.

F. CODE REQUIREMENTS:

1. ALL WORK SHALL CONFORM TO THE MINIMUM STANDARDS OF THE FOLLOWING CODES:
 - 2018 INTERNATIONAL BUILDING CODE (IBC)
 - ASCE-7-16: MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES.
 - ACI 318-14: BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE.
 - ACI 350-20: CODE REQUIREMENTS FOR ENVIRONMENTAL ENGINEERING CONCRETE STRUCTURES.
 - AISC DESIGN GUIDE 27: STRUCTURAL STAINLESS STEEL, 2013
 - ANY OTHER REGULATING AGENCIES WHICH MAY HAVE AUTHORITY OVER ANY PORTION OF THE WORK.
2. SPECIFICATIONS, CODES AND STANDARDS NOTED SHALL BE OF THE LATEST APPROVED ISSUE, INCLUDING SUPPLEMENTS, UNLESS NOTED OTHERWISE.

- 2) CONCRETE:
- A. GENERAL:
1. THE CONTRACTOR SHALL PROVIDE CAST-IN-PLACE CONCRETE, JOINTS IN CONCRETE, REINFORCEMENT STEEL AND APPURTENANT WORK, COMPLETE AND IN PLACE, IN ACCORDANCE WITH THE CONTRACT DOCUMENTS.
- B. CONCRETE – MATERIALS:
1. CEMENT SHALL BE STANDARD BRAND PORTLAND CEMENT CONFORMING TO ASTM C 150 – PORTLAND CEMENT, FOR TYPE I/II OR TYPE V.
 2. AGGREGATE SHALL BE NON-REACTIVE.
 3. WATER FOR MIXING AND CURING SHALL BE POTABLE, CLEAN, AND FREE FROM OBJECTIONABLE QUANTITIES OF SILTY ORGANIC MATTER, ALKALI, SALTS, AND OTHER IMPURITIES.
 4. ALL ADMIXTURES SHALL BE APPROVED BY THE USFWS PRIOR TO USE. IF THE USE OF AN ADMIXTURE IS PRODUCING AN INFERIOR RESULT, THE CONTRACTOR SHALL DISCONTINUE THE USE OF THE ADMIXTURE.
 5. CONCRETE PERFORMANCE CRITERIA SHALL BE PER PROJECT SPECIFICATIONS. THESE CRITERIA INCLUDE, YET ARE NOT LIMITED TO, THE FOLLOWING:
 - a. STRUCTURAL CONCRETE (INTERIOR):
 - i. MIN 28-DAY COMPRESSIVE STRENGTH, PSI: 4,500.
 - ii. MAX WATER/CEMENT RATIO, BY WEIGHT: 0.42.
 - iii. ENTRAINED AIR, PERCENT (BY VOLUME): 3-6.
- C. ALL CONCRETE WORK SHALL CONFORM TO THE LATEST EDITION OF ACI 301 AND ACI 117.
- D. ALL FOUNDATION CONCRETE SHALL BE NORMAL WEIGHT CONCRETE.
- E. CONCRETE MIX DESIGN SHALL BE ESTABLISHED IN ACCORDANCE WITH CHAPTER 5 OF ACI 318.
- F. APPROVED ADMIXTURES:
1. FLYASH PER ASTM C618
 2. AIR ENTRAINING PER ASTM C260
 3. WATER REDUCING PER ASTM C494
- E. REINFORCEMENT FOR CONCRETE:
1. ALL REINFORCING SHALL BE SUPPORTED IN FORMS SPACED WITH NECESSARY ACCESSORIES AND SHALL BE SECURELY WIRED TOGETHER IN ACCORDANCE WITH THE LATEST EDITION OF THE CRSI "MANUAL OF STANDARD PRACTICE"
 - a) DEFORMED BARS – ASTM A615, GRADE 60.
- F. MINIMUM CONCRETE CLEAR COVER OVER REINFORCEMENT FOR HYDRAULIC AND ENVIRONMENTAL STRUCTURES:
1. CAST-IN-PLACE CONCRETE
 - a) CONCRETE CAST AGAINST EARTH = 3"
 - b) ALL OTHER CONCRETE = 2"
- G. REINFORCEMENT BENDS AND LAPS, UNLESS OTHERWISE NOTED, SHALL SATISFY THE FOLLOWING MINIMUM REQUIREMENTS:

TYPICAL LAP SPLICE LENGTHS IN INCHES, PER ACI 318/350					
BAR SIZE (#)	BAR POSITION	CLASS B LAP LENGTH		Ld (DEVELOPMENT LENGTH)	
		SPACING >= 6"	SPACING < 6"	SPACING >= 6"	SPACING < 6"
3	BOTTOM	16	16	12	12
	TOP	16	16	12	12
4	BOTTOM	16	18	12	14
	TOP	19	23	14	18
5	BOTTOM	18	26	14	20
	TOP	23	34	18	26
6	BOTTOM	21	35	17	27
	TOP	28	46	21	35
7	BOTTOM	31	51	24	40
	TOP	40	67	31	51
8	BOTTOM	35	59	27	45
	TOP	46	76	35	59
9	BOTTOM	44	66	34	51
	TOP	56	86	44	66
10	BOTTOM	52	73	40	56
	TOP	68	95	52	73
11	BOTTOM	62	80	48	62
	TOP	80	104	62	80

- NOTES:
1. FOR GRADE 60 REINFORCING STEEL BARS.
 2. FOR CONCRETE COMPRESSIVE STRENGTH f'c=4,500 PSI.
 3. TOP BARS ARE HORIZONTAL BARS WITH MORE THAN 12" OF CONCRETE CAST BELOW THE BARS.
 4. ALL REINFORCING HOOKS SHALL BE PER ACI STANDARDS.

H. FORMWORK: DESIGN, ERECT, SUPPORT, BRACE AND MAINTAIN FORMWORK TO SUPPORT VERTICAL, LATERAL, STATIC AND DYNAMIC LOADS THAT MIGHT BE APPLIED UNTIL STRUCTURE CAN SUPPORT SUCH LOADS.

3) STRUCTURAL AND MISCELLANEOUS STEEL:

- A. ALL STRUCTURAL STEEL WORK SHALL CONFORM TO THE LATEST EDITION OF THE AISC SPECIFICATIONS.
- B. WELDS: PROVIDE 70ksi LOW HYDROGEN ELECTRODE OR PROCESS IN ACCORDANCE WITH AWS A5.1.
- C. EPOXY BOLT OR EXPANSION BOLT SUBSTITUTIONS FOR EMBEDDED BOLTS IS PROHIBITED WITHOUT WRITTEN CONSENT FROM THE ENGINEER.
- D. ALL STRUCTURAL STEEL SHALL BE DETAILED, FABRICATED, AND ERECTED IN ACCORDANCE WITH THE AISC CODE OF STANDARD PRACTICE, EXCEPT AS MODIFIED IN THESE NOTES AND THE PROJECT SPECIFICATIONS.

CONCEPT DESIGN
NOT FOR CONSTRUCTION



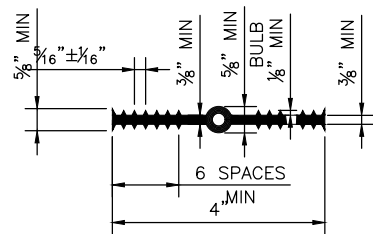
VERIFY SCALE

THIS BAR IS ONE INCH ON ORIGINAL DRAWING

0" ————— 1"

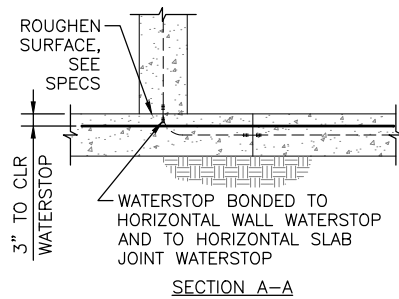
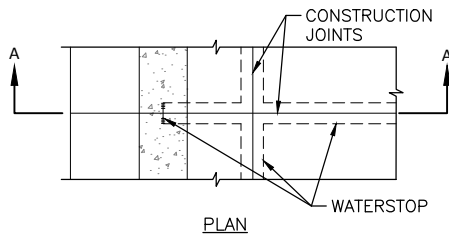
ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET

	07/28/23	95% DESIGN SUBMITTAL	RH
	05/05/23	65% DESIGN SUBMITTAL	JB
	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
GENERAL STRUCTURAL NOTES			GS001
DESIGNED KH	DRAWN DD	CHECKED JB	DATE 07/28/23
DRAWING NO.			



4" CENTER-BULB WATERSTOP

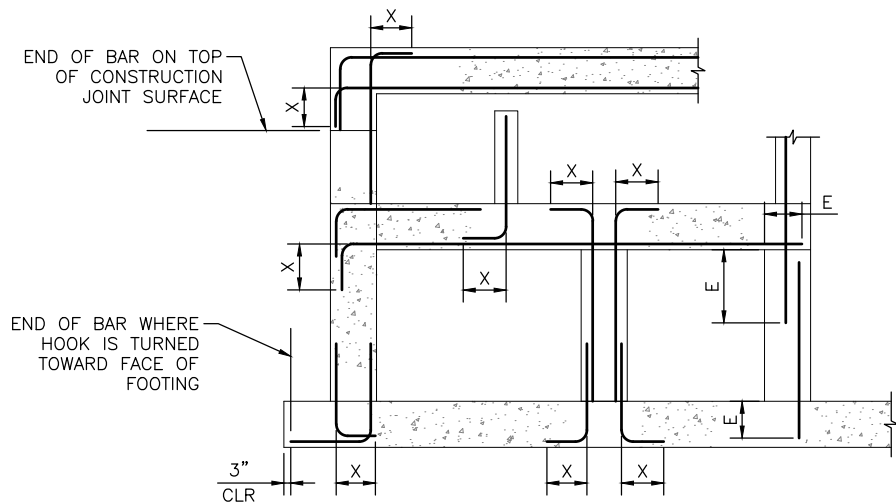
SCALE: NTS



- NOTES:
1. CONSTRUCTION JOINTS PASSING THROUGH VARIOUS MEMBERS OF A WATER RETAINING STRUCTURE SHALL BE SEALED WITH WATERSTOPS BONDED TOGETHER, SO AS TO PROVIDE A CONTINUOUS WATERTIGHT JOINT.

CONSTRUCTION JOINT (WALL TO SLAB)

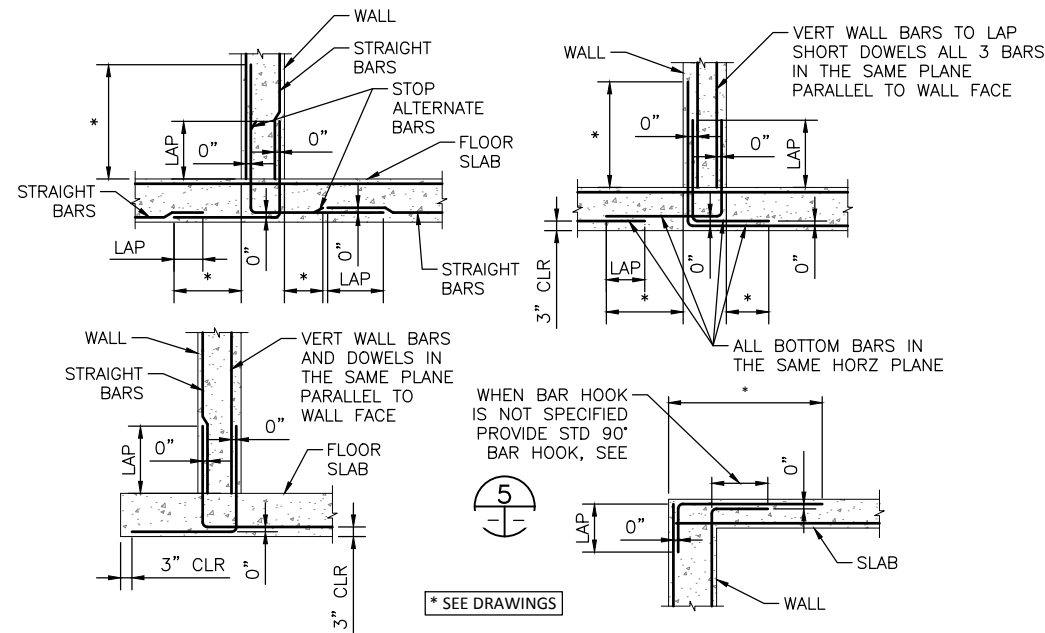
SCALE: NTS



LENGTH (*)		
BAR SIZE	HOOK X	EMBEDMENT E
#3	6"	12" (16")
#4	8"	12" (16")
#5	10"	15" (20")
#6	12"	22" (28")
#7	14"	37" (48")
#8	16"	48" (62")
#9	19"	61" (79")
#10	22"	77" (100")
#11	24"	95" (123")

* USE LENGTH IN PARENTHESIS FOR WALL HORIZONTAL REBARS AND SLAB BARS WITH 12" OR MORE OF FRESH CONCRETE UNDERNEATH

- NOTES:
1. USE LAP LENGTHS AS DETERMINED FROM THESE TABLES UNLESS SHOWN OTHERWISE.
 2. THE TABLES SHOWN ARE FOR $f'c=4500\text{psi}$, $f_y=60,000\text{psi}$, 1.5" MIN CONCRETE COVER AND 3" MIN BAR SPACING.
 3. ALL DOWEL BARS SHALL EXTEND AN EMBEDMENT LENGTH E INTO ANOTHER MEMBER OR ACROSS A CONSTRUCTION JOINT UNLESS SHOWN TO SPLICE WITH OTHER BARS OR TO EXTEND TO THE FAR FACE OF THE MEMBER AND END WITH A STANDARD HOOK.



STANDARD 90° BAR HOOKS, EMBEDMENT LENGTHS AND LAP LENGTHS

SCALE: NTS



STANDARD FOR REINFORCEMENT BAR DETAILING

SCALE: NTS



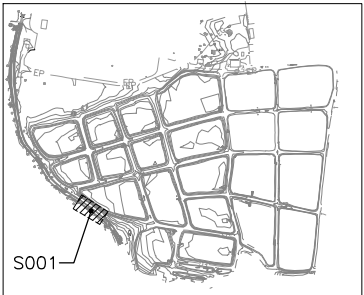
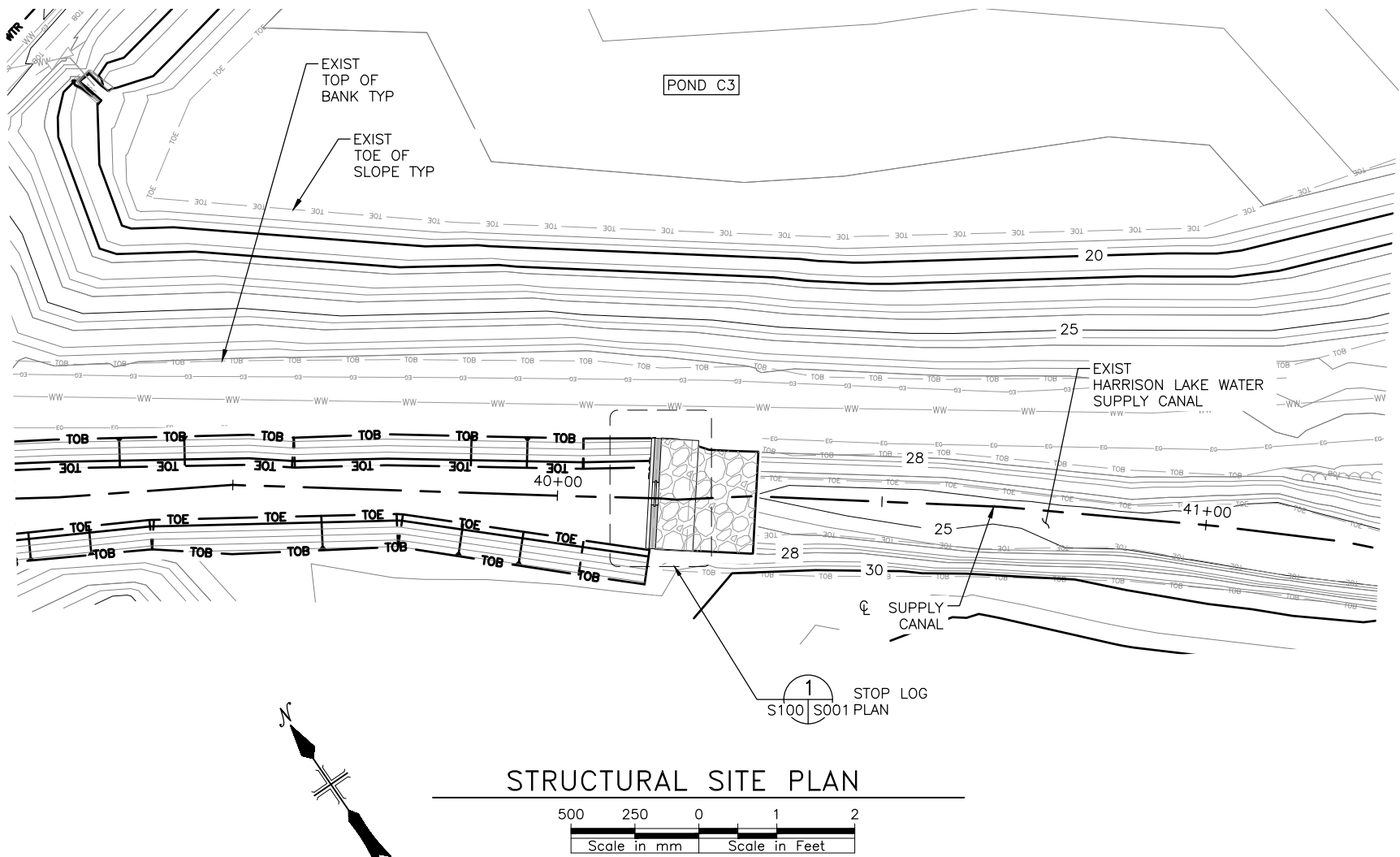
CONCEPT DESIGN
NOT FOR CONSTRUCTION



VERIFY SCALE
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0" 1"
ADJUST SCALES ACCORDINGLY, IF NOT ONE INCH ON THIS SHEET

△	07/28/23	95% DESIGN SUBMITTAL	RH
△	05/05/23	65% DESIGN SUBMITTAL	JB
△	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
STANDARD STRUCTURAL DETAILS			GS002
DESIGNED KH	DRAWN DD	CHECKED JB	DATE 05/05/23
DRAWING NO.			

- SHEET NOTES:
- CONTRACTOR SHALL REPAIR EXISTING CANAL BANKS AND DREDGE BOTTOM OF CHANNEL. AS REQUIRED PER CIVIL SHEETS.



KEY PLAN

CONCEPT DESIGN
NOT FOR CONSTRUCTION



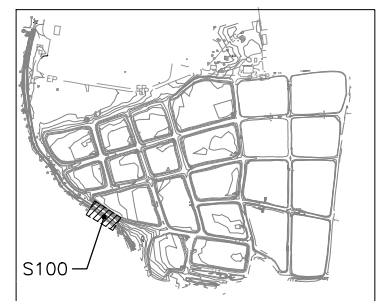
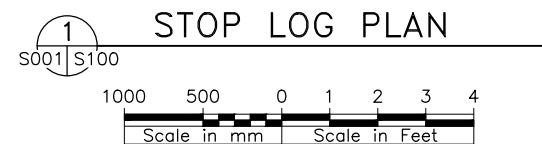
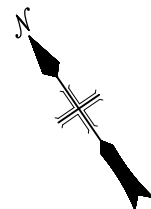
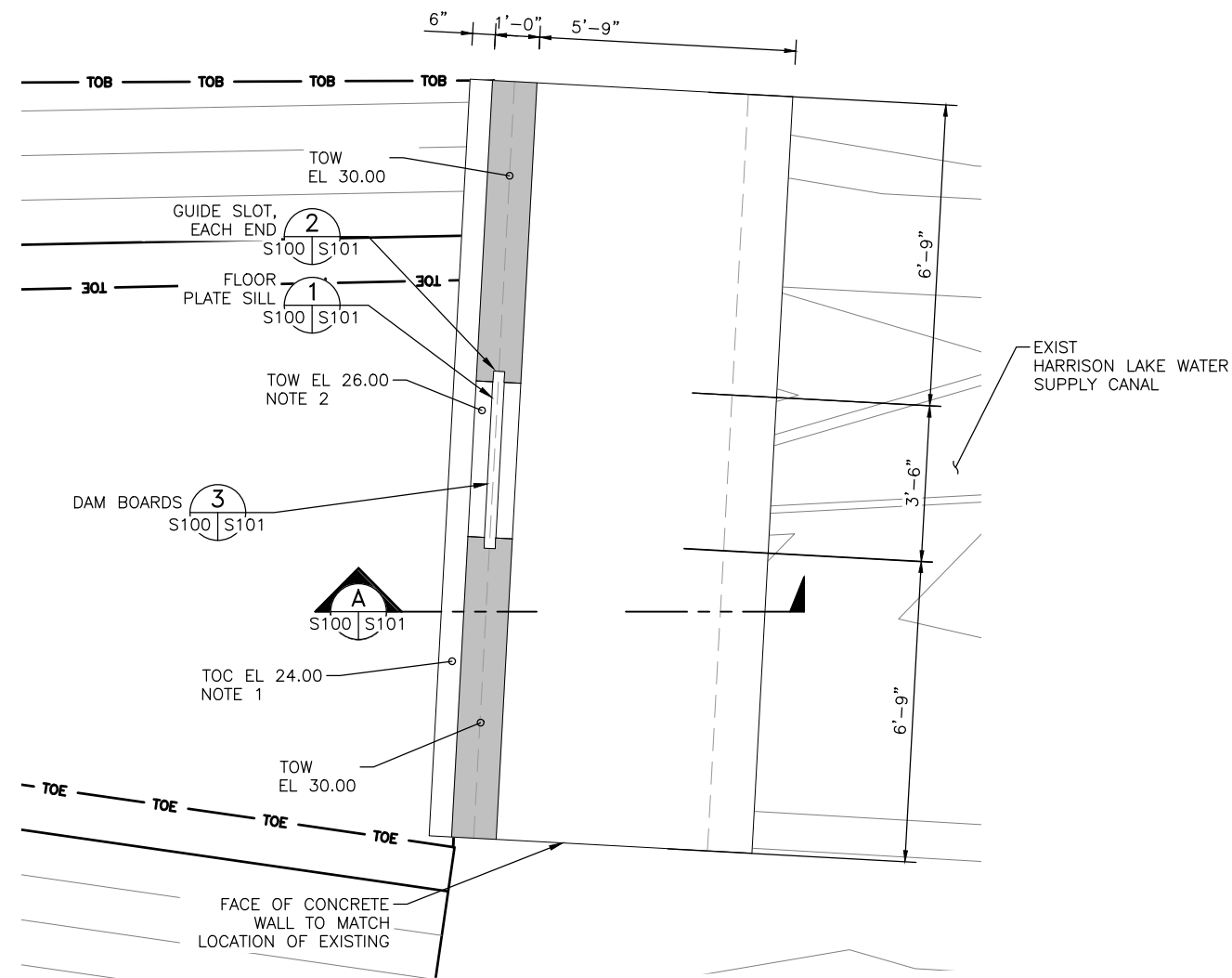
VERIFY SCALE
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ORIGINAL DRAWING
0" 1"
ADJUST SCALES
ACCORDINGLY, IF
NOT ONE INCH
ON THIS SHEET

DESIGNED	DRAWN	CHECKED	DATE	DRAWING NO.
KH	DD	JB	07/28/23	----

REV.	DATE	DESCRIPTION	BY
A	07/28/23	95% DESIGN SUBMITTAL	RH
B	05/05/23	65% DESIGN SUBMITTAL	JB
A	10/28/22	35% SCHEMATIC DESIGN	JB

HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
STRUCTURAL SITE PLAN			S001

- SHEET NOTES:
1. TOP OF FOOTING ELEVATION TO BE 1'-0" BELOW LOWEST ADJACENT GRADE. FIELD VERIFY TOP OF FOOTING ELEVATION AND WALL HEIGHT WITH FIELD CONDITIONS.
 2. TOP OF WALL ELEVATION TO BE 1'-0" ABOVE LOWEST ADJACENT GRADE. FIELD VERIFY TOP OF WALL ELEVATION WITH FIELD CONDITIONS.
 3. FIELD VERIFY WALL LENGTHS WITH FIELD CONDITIONS. WALLS TO EXTEND INTO CANAL BANK A MINIMUM OF 1'-0" TO PREVENT WATER SEEPAGE AROUND WALLS.



KEY PLAN

CONCEPT DESIGN
NOT FOR CONSTRUCTION



VERIFY SCALE
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ORIGINAL DRAWING
0" 1"
ADJUST SCALES
ACCORDINGLY, IF
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ON THIS SHEET

DESIGNED	DRAWN	CHECKED	DATE	DRAWING NO.
KH	DD	JB	07/28/23	----

REV.	DATE	DESCRIPTION	BY
A	07/28/23	95% DESIGN SUBMITTAL	RH
B	05/05/23	65% DESIGN SUBMITTAL	JB
A	10/28/22	35% SCHEMATIC DESIGN	JB

HARRISON LAKE NFH – WATER SUPPLY REHABILITATION			
STOP LOG PLAN			
S100			



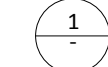
NOTE:
316 SST MAY BE SUBSTITUTED WITH EXTRUDED OR
BUILT UP ALUMINUM (COATED PER SPECIFICATIONS)
AT CONTRACTOR'S DISCRETION.



SCALE: NTS



SCALE: NTS



EL= SEE PLAN

DAM BOARD SCHEDULE

LENGTH (L)

BOARDS

3'-10''

8

NOTES:

1. LENGTH (L) PER LOCATION — SEE TABLE.
2. DAM BOARD LENGTH TO BE FIELD VERIFIED BEFORE FABRICATION.
3. NUMBER OF DAM BOARDS TO BE FIELD VERIFIED BEFORE FABRICATION.

SCALE: NTS



- NOTES:
1. TOP OF FOOTING ELEVATION TO BE 1'-0" BELOW LOWEST ADJACENT GRADE. FIELD VERIFY TOP OF FOOTING ELEVATION AND WALL HEIGHT WITH FIELD CONDITIONS.

SCALE: NTS



CONCEPT DESIGN
NOT FOR CONSTRUCTION

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	05/05/23	65% DESIGN SUBMITTAL	JB
	10/28/22	35% SCHEMATIC DESIGN	JB
REV.	DATE	DESCRIPTION	BY
HARRISON LAKE NFH - WATER SUPPLY REHABILITATION			
STOP LOG SECTION AND DETAILS			S101
DESIGNED KH	DRAWN DD	CHECKED JB	DATE 07/28/23
DRAWING NO.			-----

SHEET ____ OF ____



APPENDIX 2:

Photographs

Water Canal and Existing Water Control Structure



Photo #1: View of access and water canal U/S of WCS.



Photo #2: View of bank and canal U/S of WCS at Water Intake.



Photo #3: View of U/S side of existing WCS to be replaced.



Photo #4: View of D/S side of existing WCS to be replaced.